SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

AGILE DEVELOPMENT PROCESS AND DEVOPS

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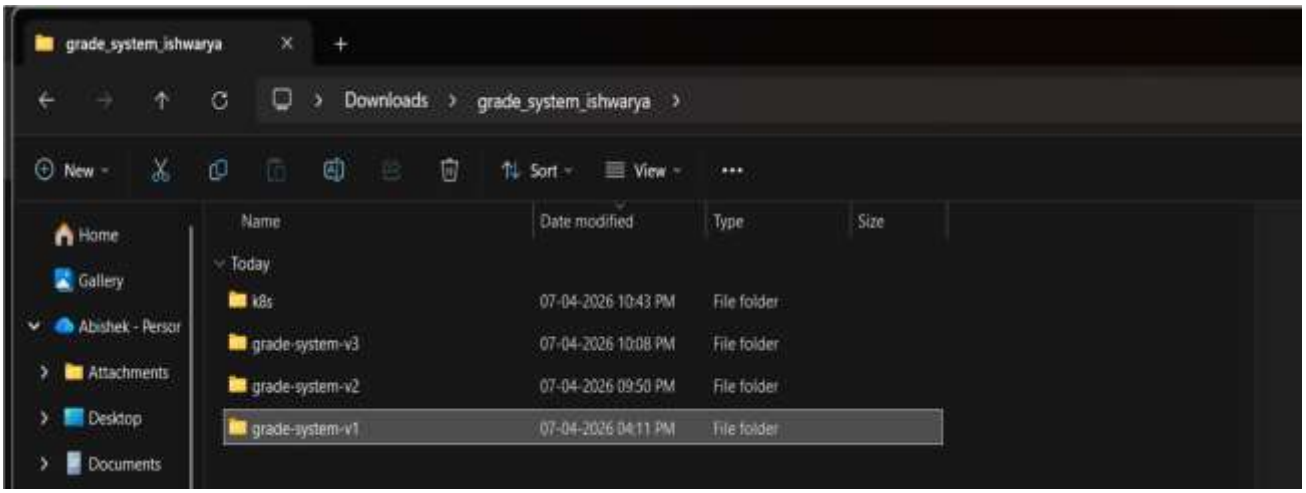
REG NO: 23MIS0028

COURSE CODE: ISWE406P

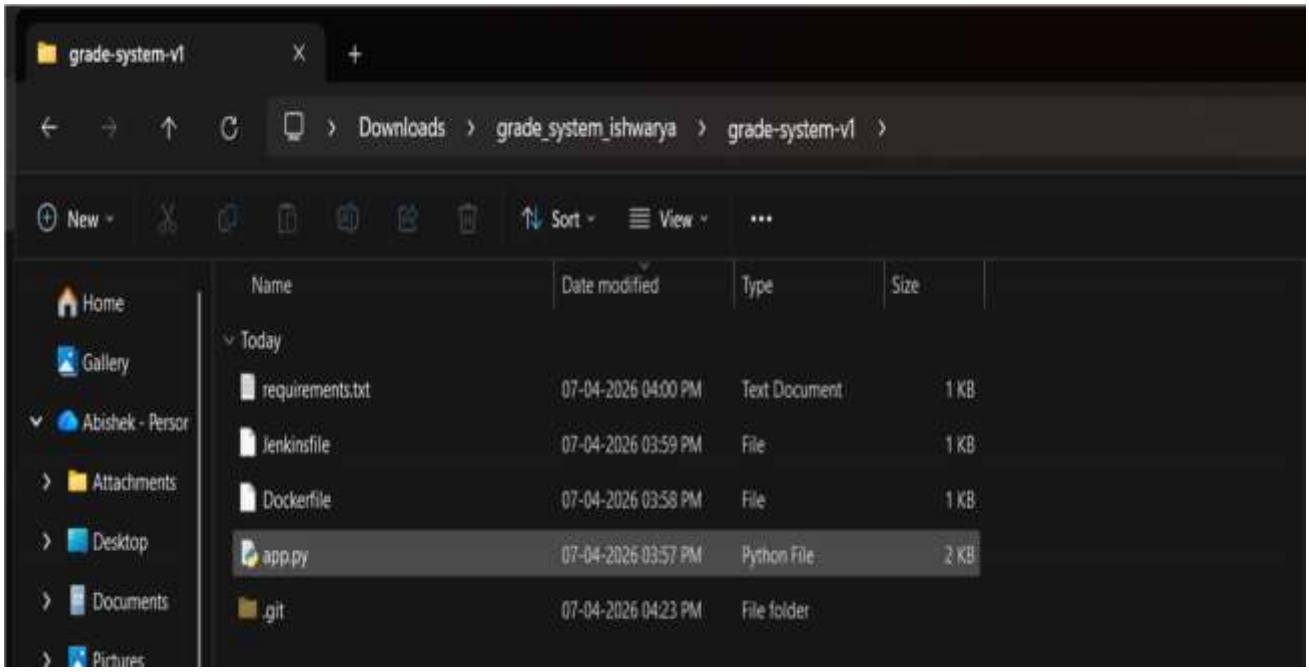
FACULTY NAME: M.RAMALINGAM

GRADE SYSTEM

Create a new folder named grade-system-v1 and open it in VS Code.



Inside it, create these 4 files:



All four file are :

App.py for v1:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

HTML = """
<!DOCTYPE html>
<html>
<head>
    <title>Grade System V1</title>
</head>
<body>
    <h2>Grade System - V1</h2>
    <h3>Enter Marks</h3>

    <form method="post">
        <label>Subject 1 Marks:</label>
        <input type="number" name="sub1" required><br><br>

        <label>Subject 2 Marks:</label>
        <input type="number" name="sub2" required><br><br>

        <label>Subject 3 Marks:</label>
        <input type="number" name="sub3" required><br><br>

        <button type="submit">Submit</button>
    </form>

    {% if sub1 is not none %}
        <h3>Entered Marks</h3>
        <p>Subject 1: {{ sub1 }}</p>
        <p>Subject 2: {{ sub2 }}</p>
        <p>Subject 3: {{ sub3 }}</p>
    {% endif %}
</body>
</html>
"""
```

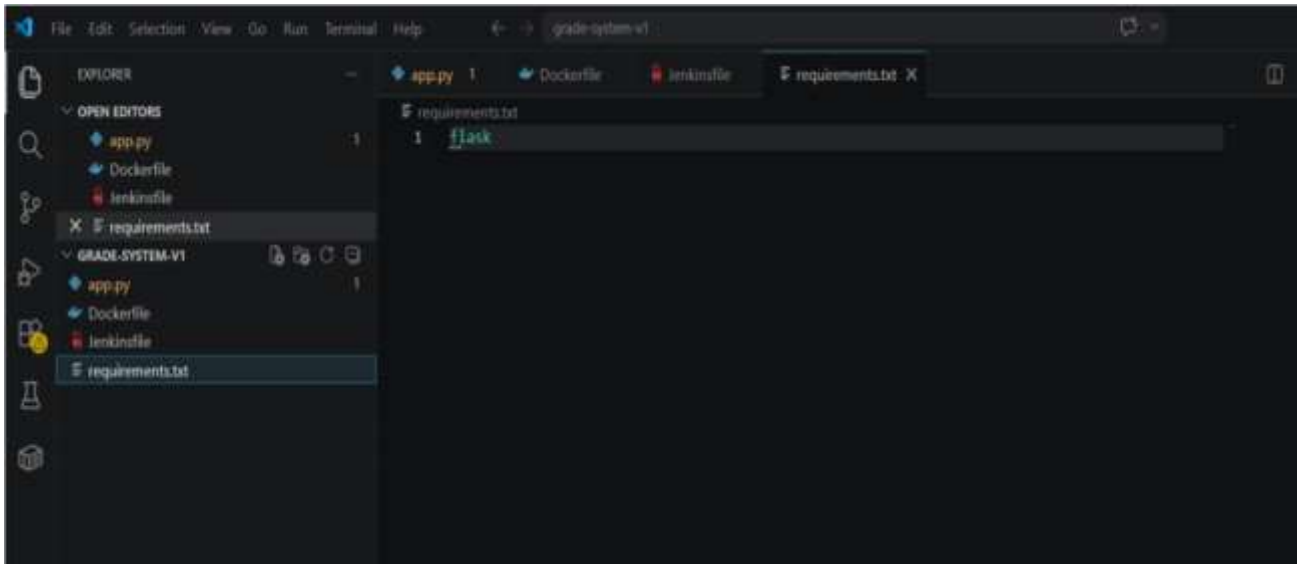
Jenkinsfile for v1 :

```
pipeline {
    agent any

    environment {
        IMAGE_NAME = "ishwarya21/grade-system"
        IMAGE_TAG = "v1"
    }

    stages {
        stage('Build Docker Image') {
            steps {
                bat 'docker build -t %IMAGE_NAME%:%IMAGE_TAG% .'
            }
        }

        stage('Push Docker Image') {
            steps {
                withCredentials([usernamePassword(credentialsId: 'dockerhub-cred-ishu', usernameVariable: 'DOCKER_USER', passwordVariable: 'DOCKER_PASS')]) {
                    bat 'docker login -u %DOCKER_USER% -p %DOCKER_PASS%'
                    bat 'docker push %IMAGE_NAME%:%IMAGE_TAG%'
                }
            }
        }
    }
}
```

Dockerfile for v1:

```
FROM python:3.10-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

EXPOSE 5000

CMD ["python", "app.py"]
```

Requirement.txt for v1:



Now push it into github repository:

Create a new repository called grade-system-v1 and push it through

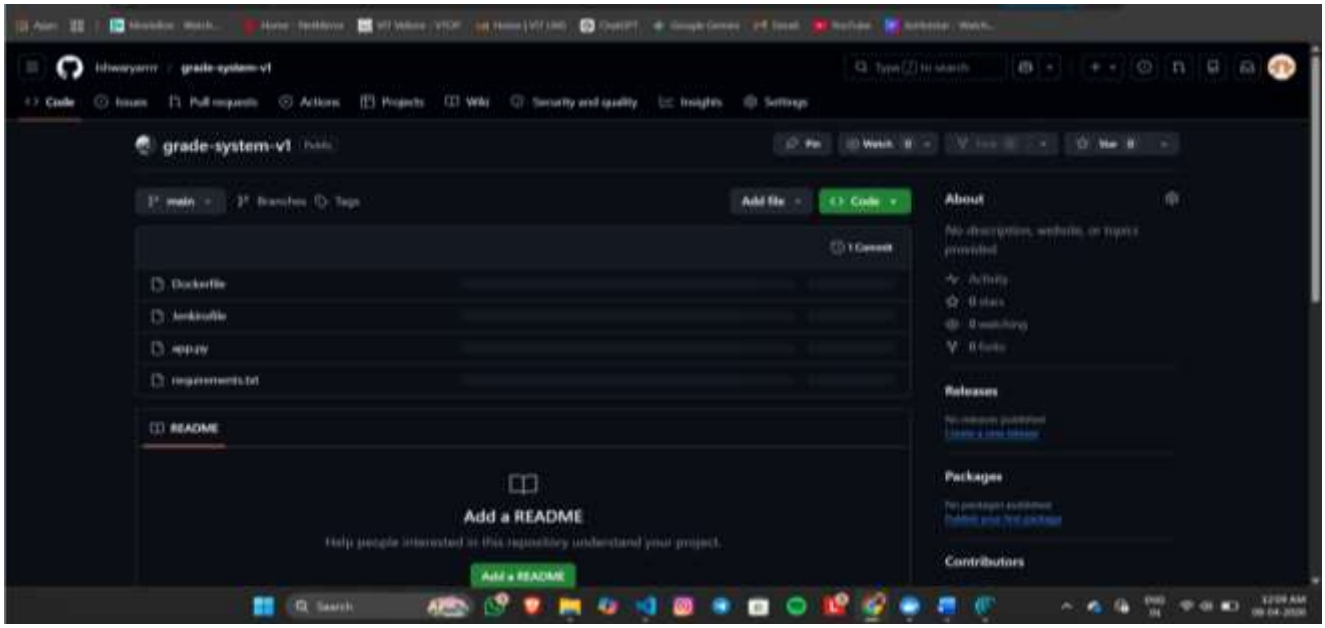
git init git add

git commit -m "v1 - enter marks"

git branch -M main
git remote add origin

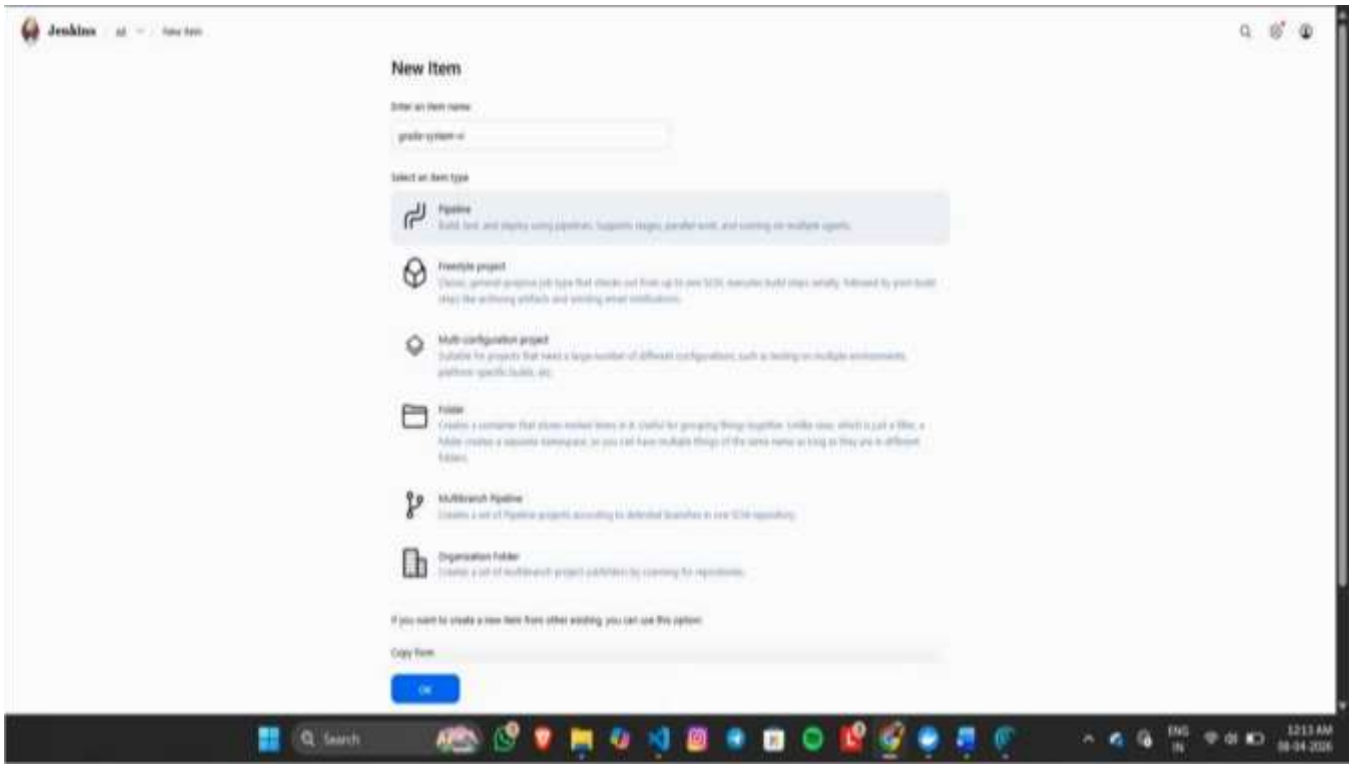
https://github.com/Ishwaryarrrr/grade-system-

v1.git git push -u origin main



Now open Jenkins :

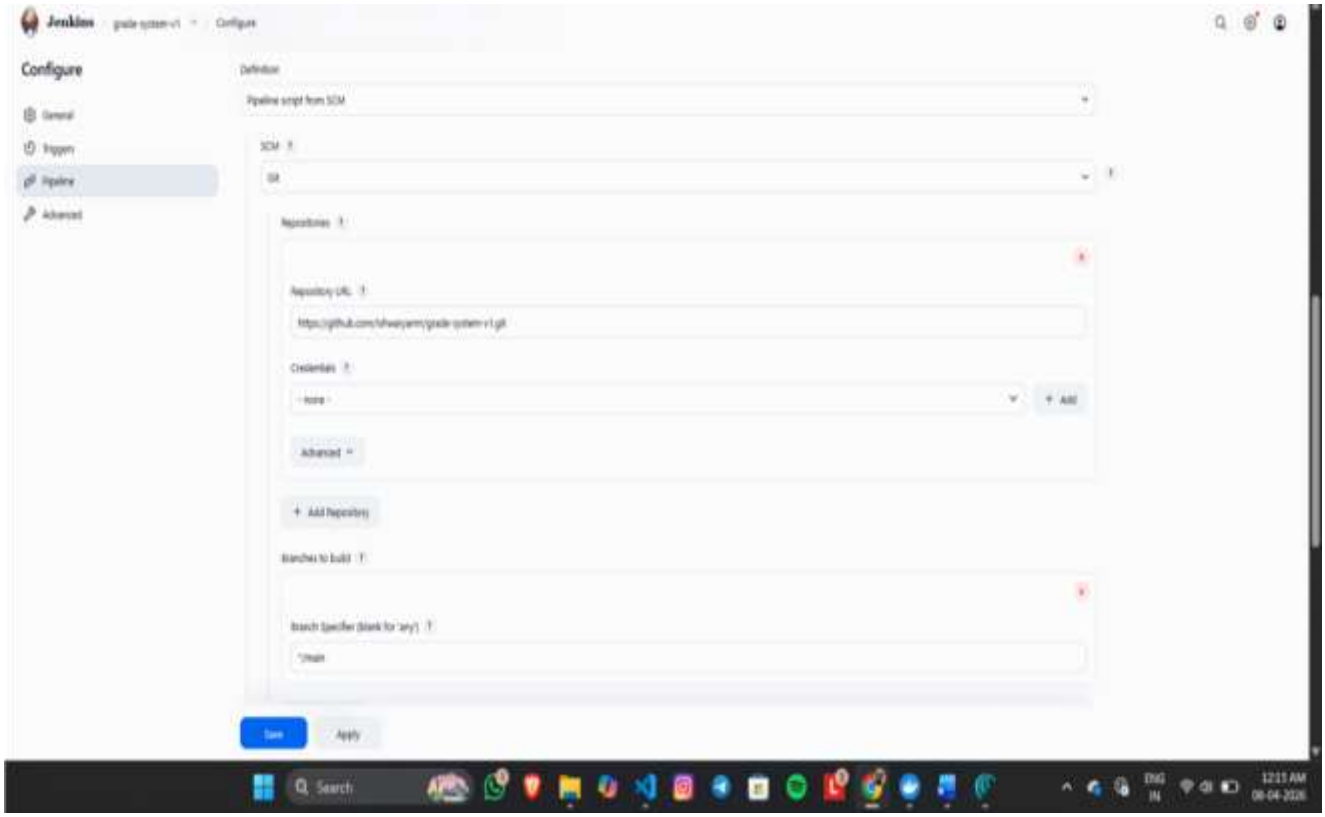
.create new item called garde system v1



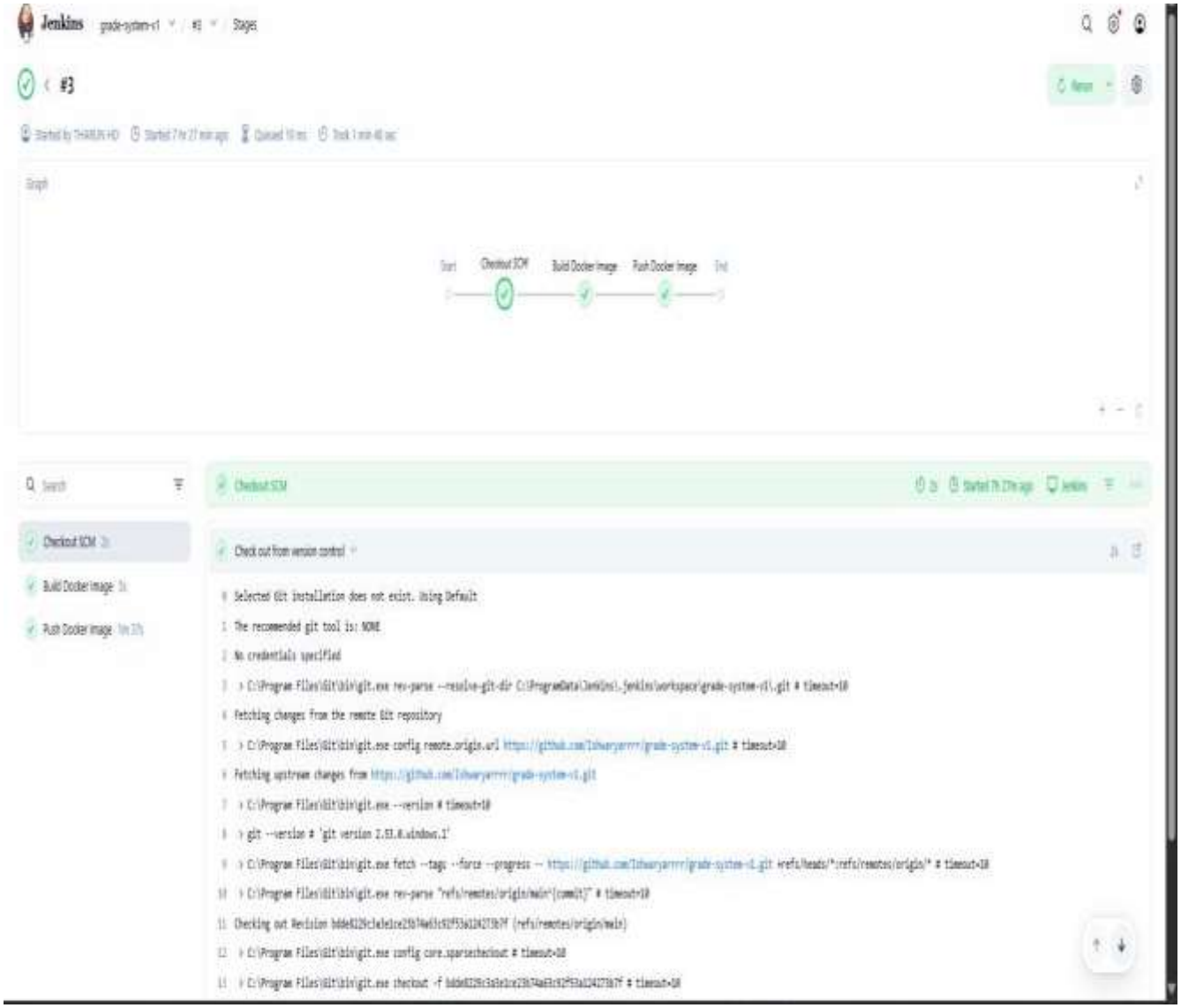
Scroll to Pipeline section.

Set:

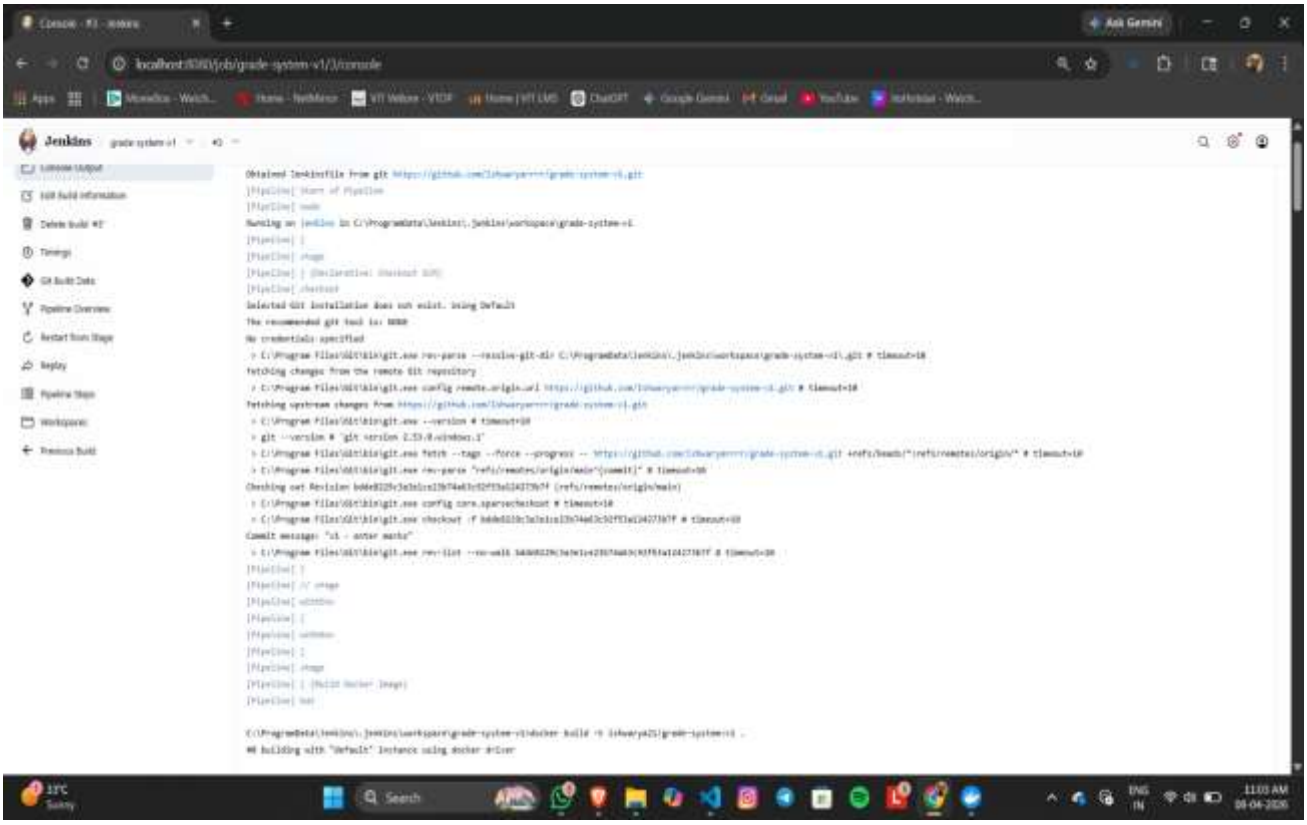
- **SCM =** Git
- **Repository URL =** https://github.com/Ishwaryarrrr/grade-system-v1.git
- **Branch Specifier** = */main



Click: Build Now

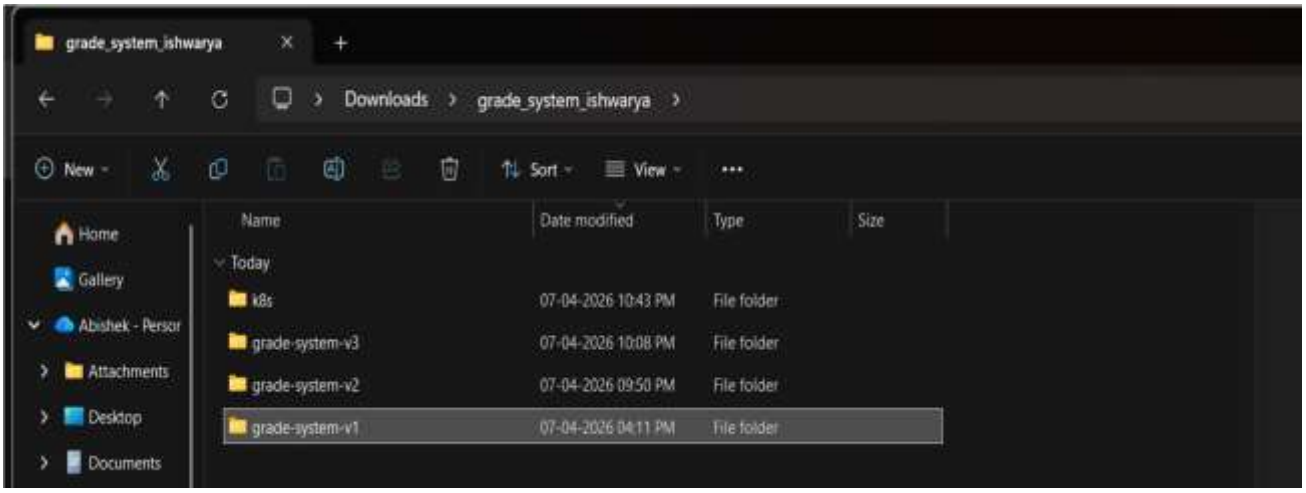


Console Output:

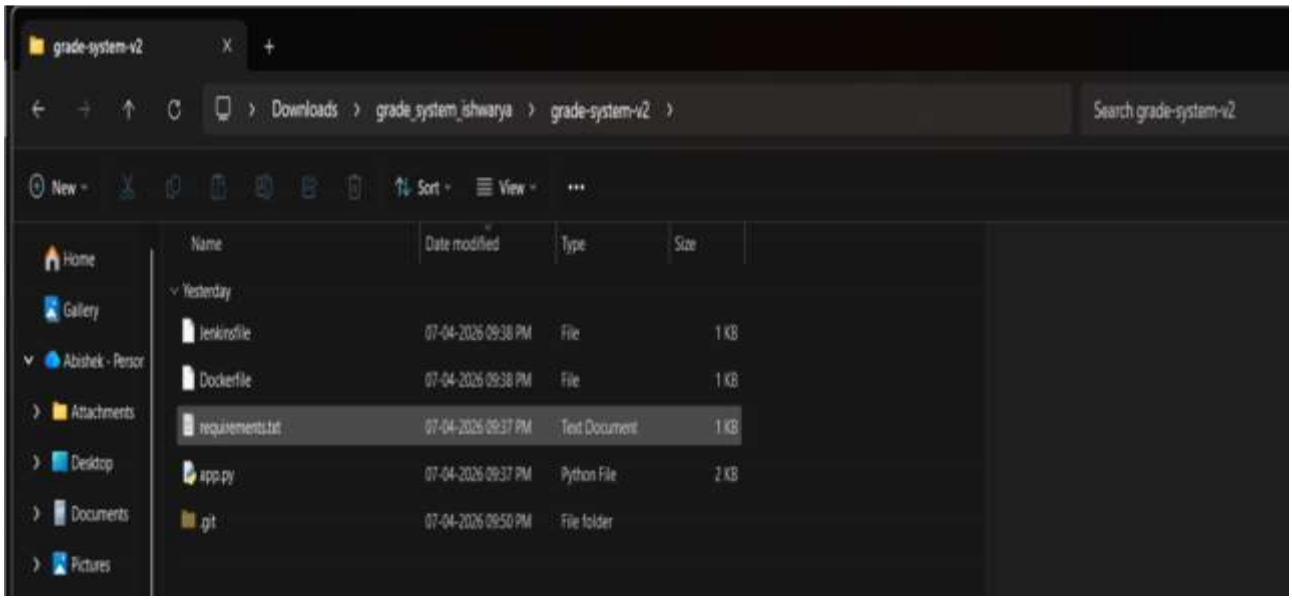


Similarly do it for v2

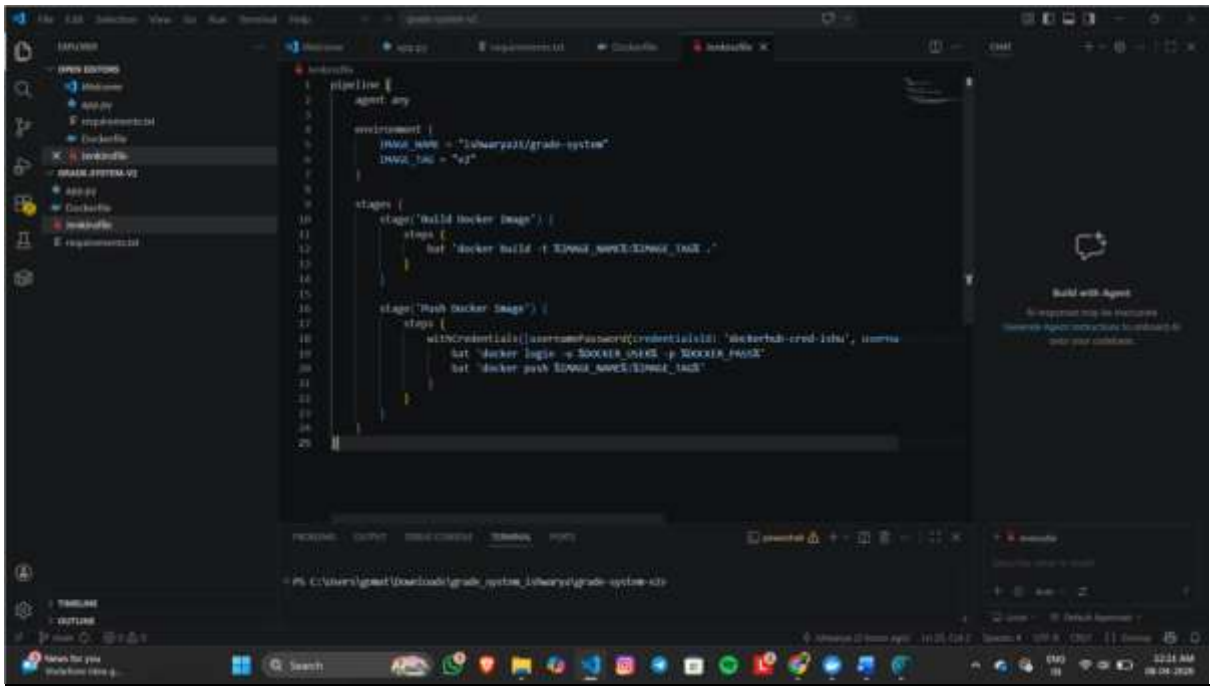
Create a new folder named grade-system-v2 and open it in VS Code.



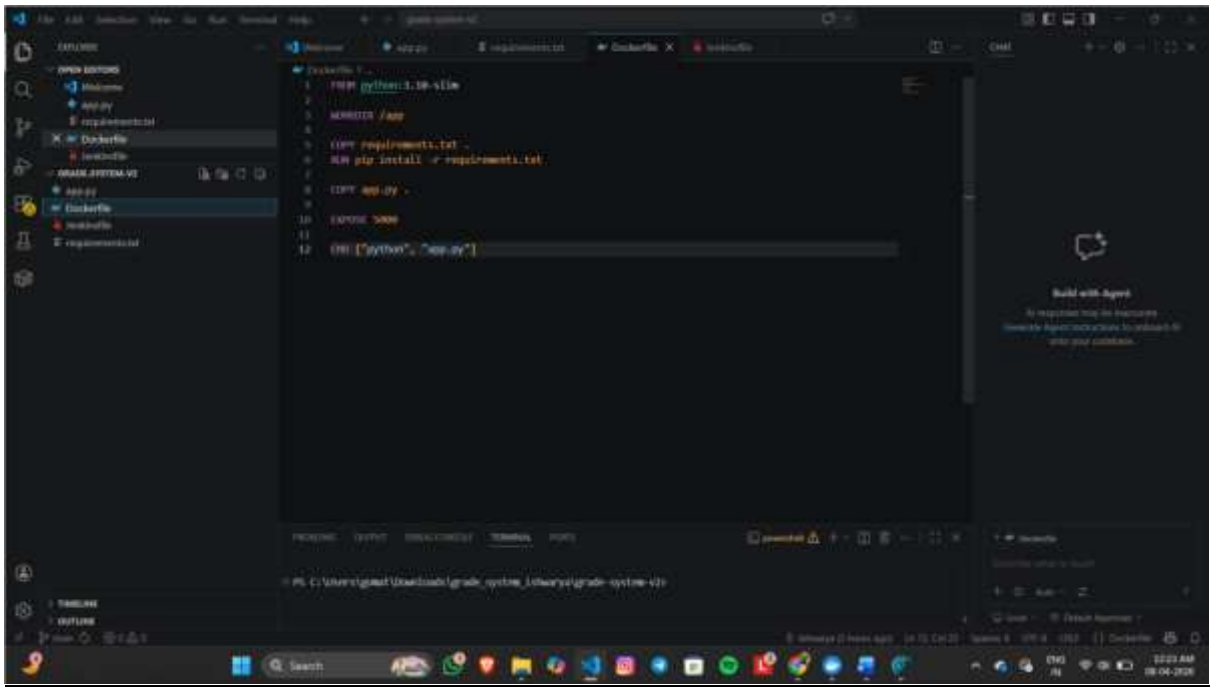
In the v2.. it should present all 4 file they are as follows:



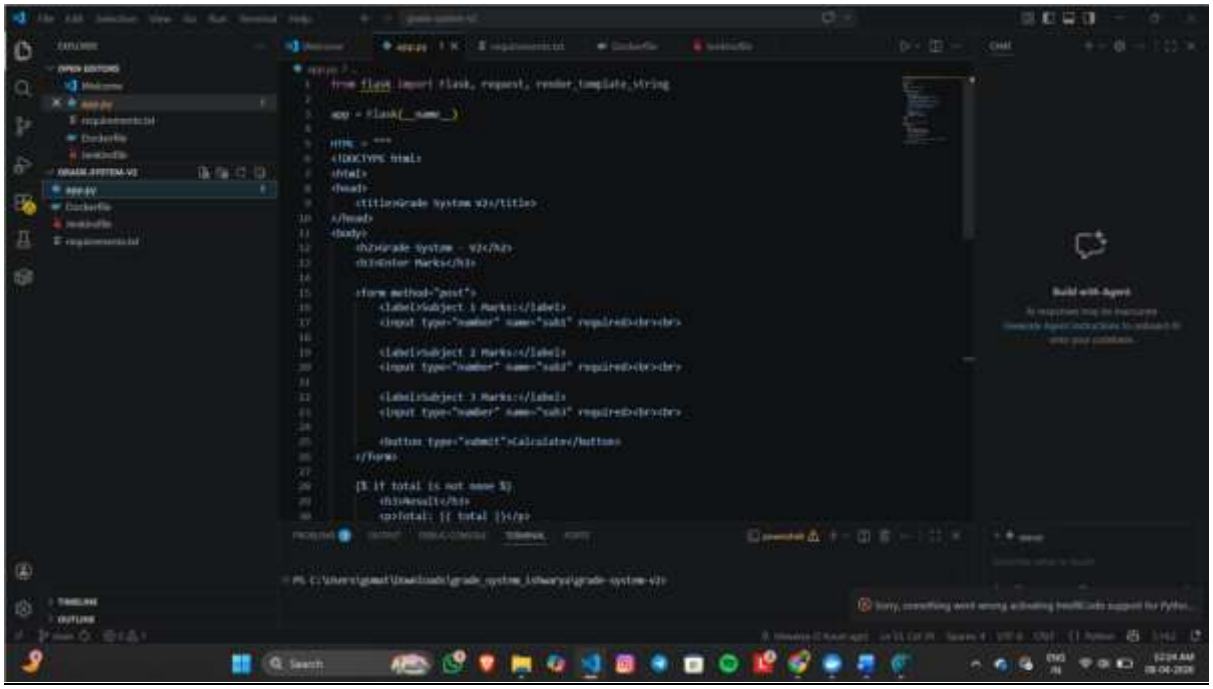
Jenkins file



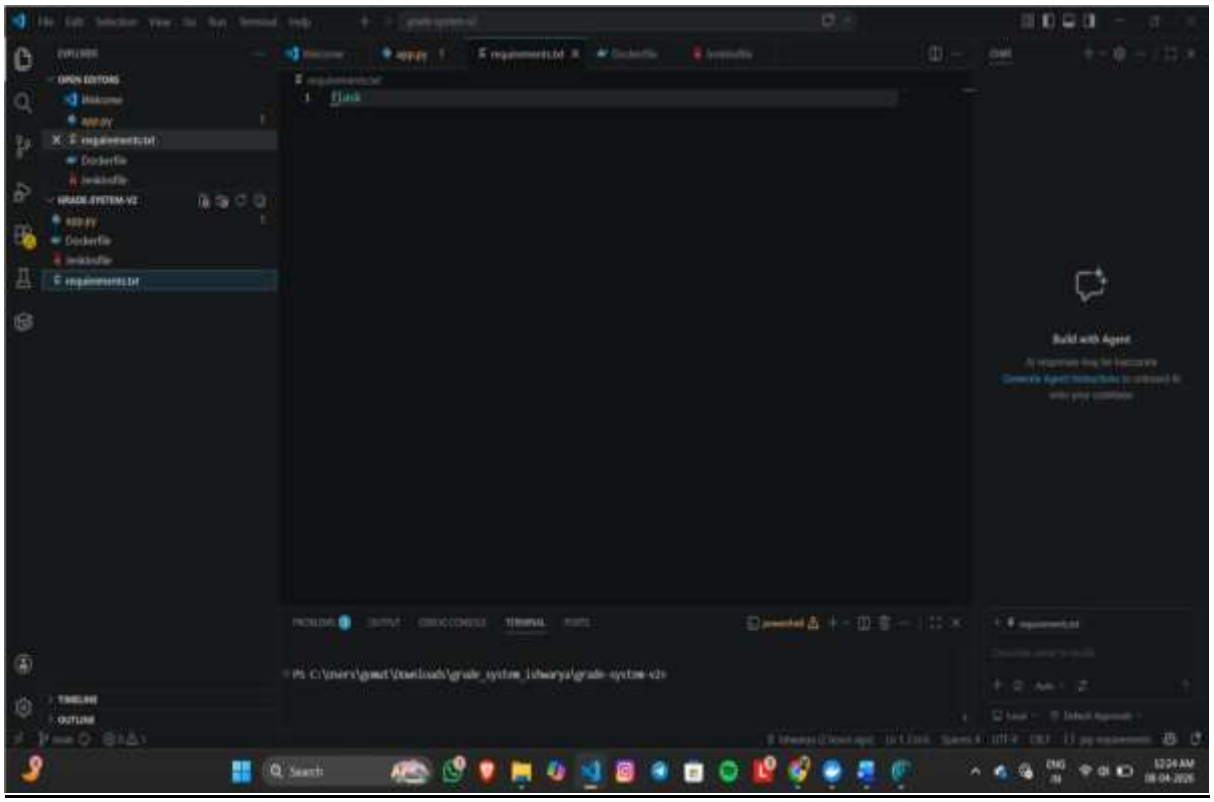
<u>Docker File:</u>



App.py:



Requirements.txt:



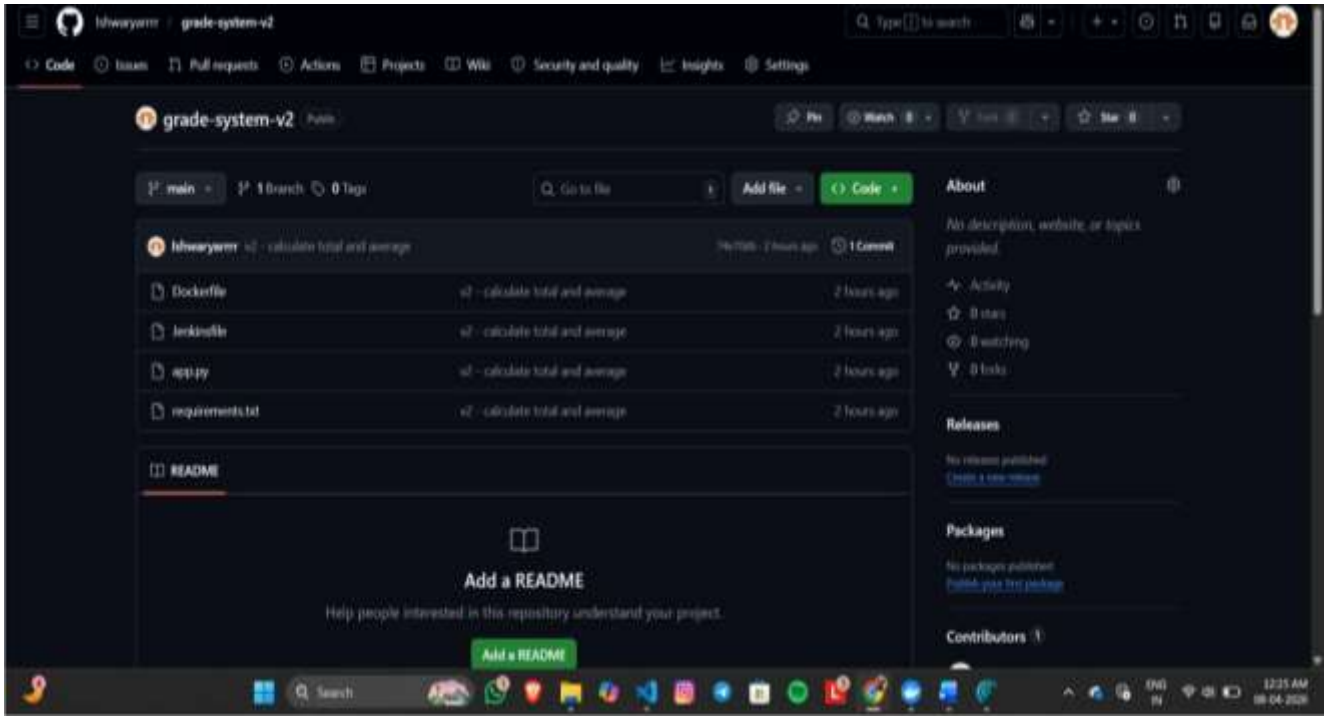
Now push it into github repository:

Create a new repository called grade-system-v1 and push it through

git init git add .

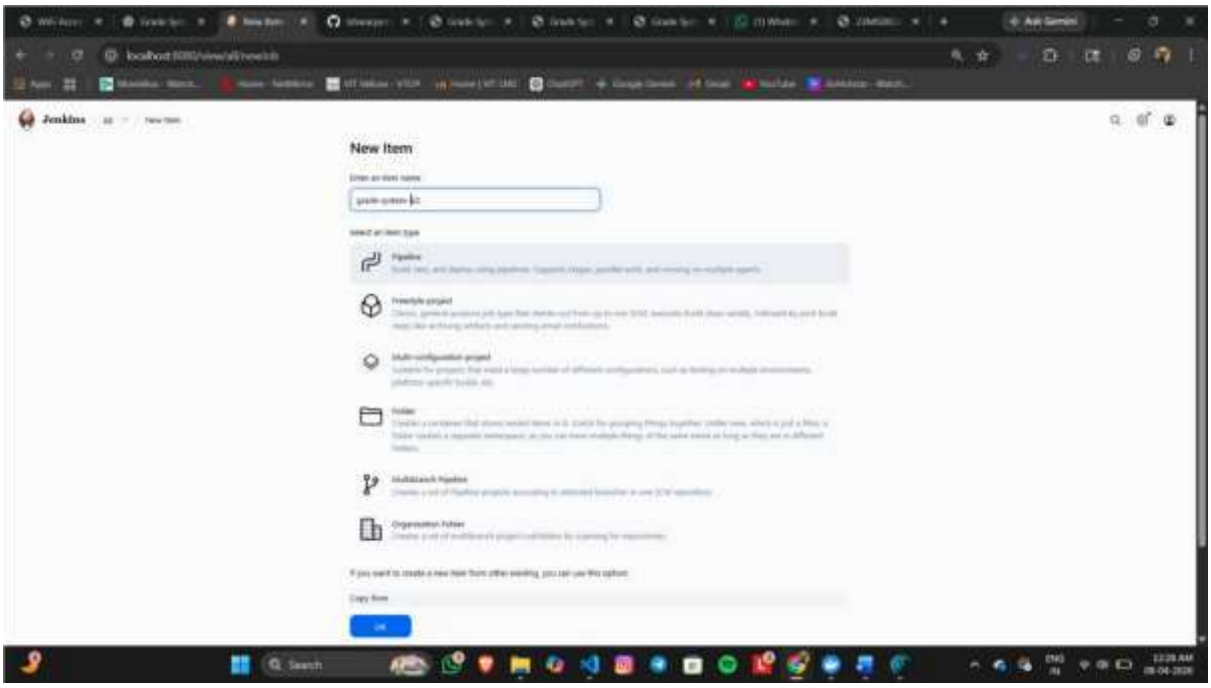
git commit -m "v2 - calculate total and average" git branch -M main

git remote add origin https://github.com/Ishwaryarrrr/grade-system-v2.git git push -u origin main



Now open Jenkins :

.create new item called garde system v2:



Scroll to Pipeline section.

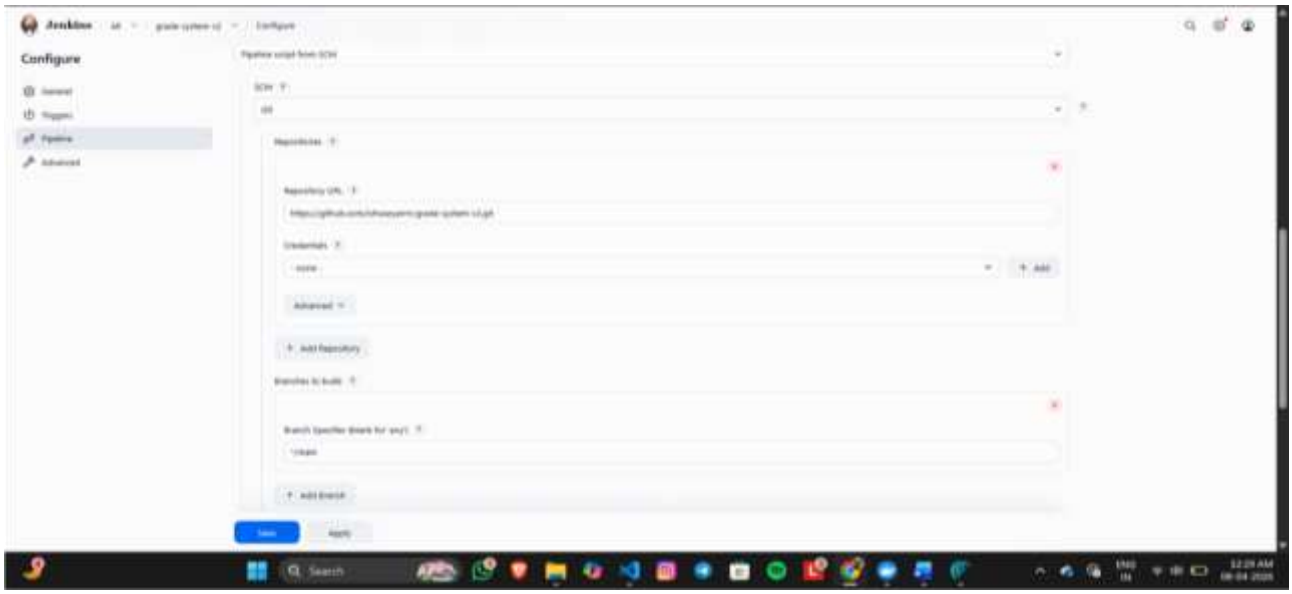
Set:

- **SCM =** Git

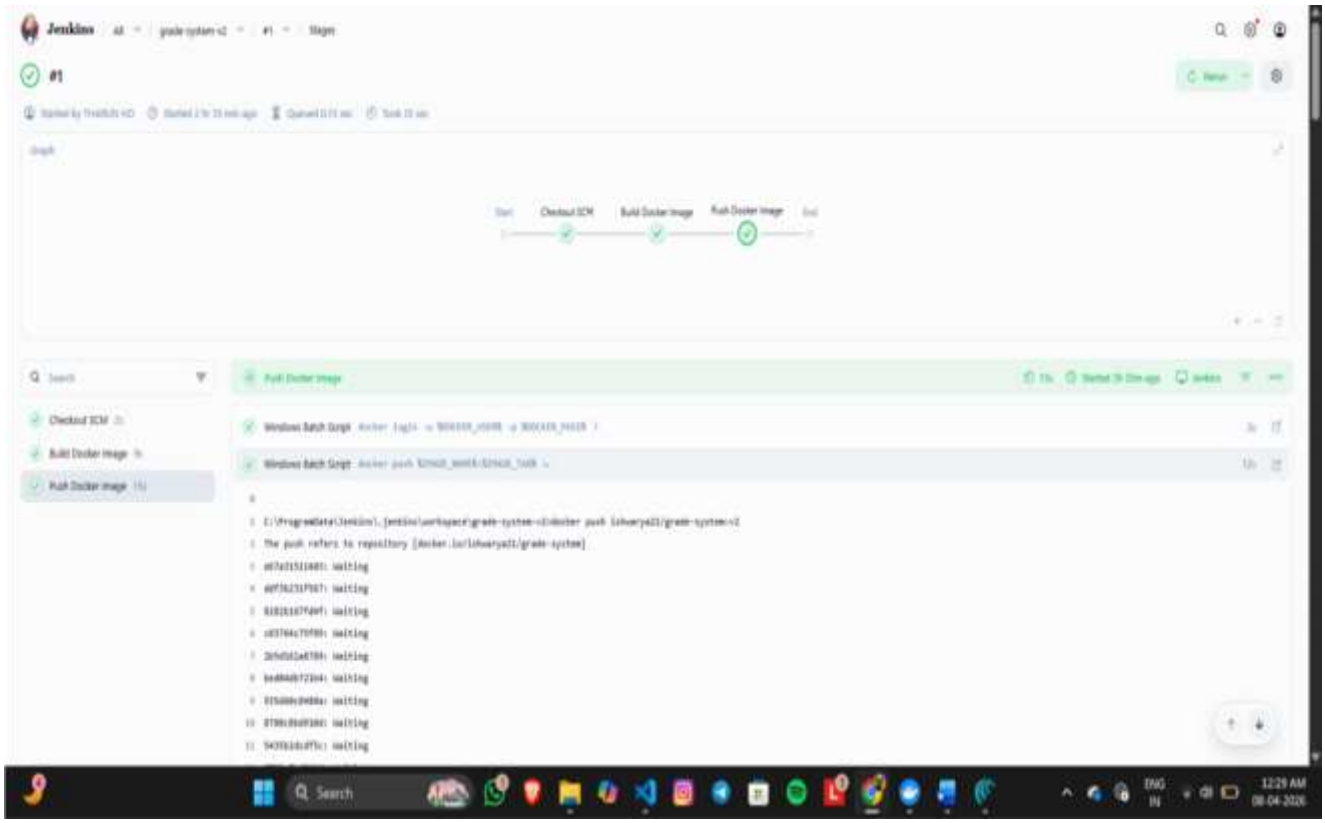
  Repository URL =

  https://github.com/Ishwaryarrrr/grade-system-v2.git

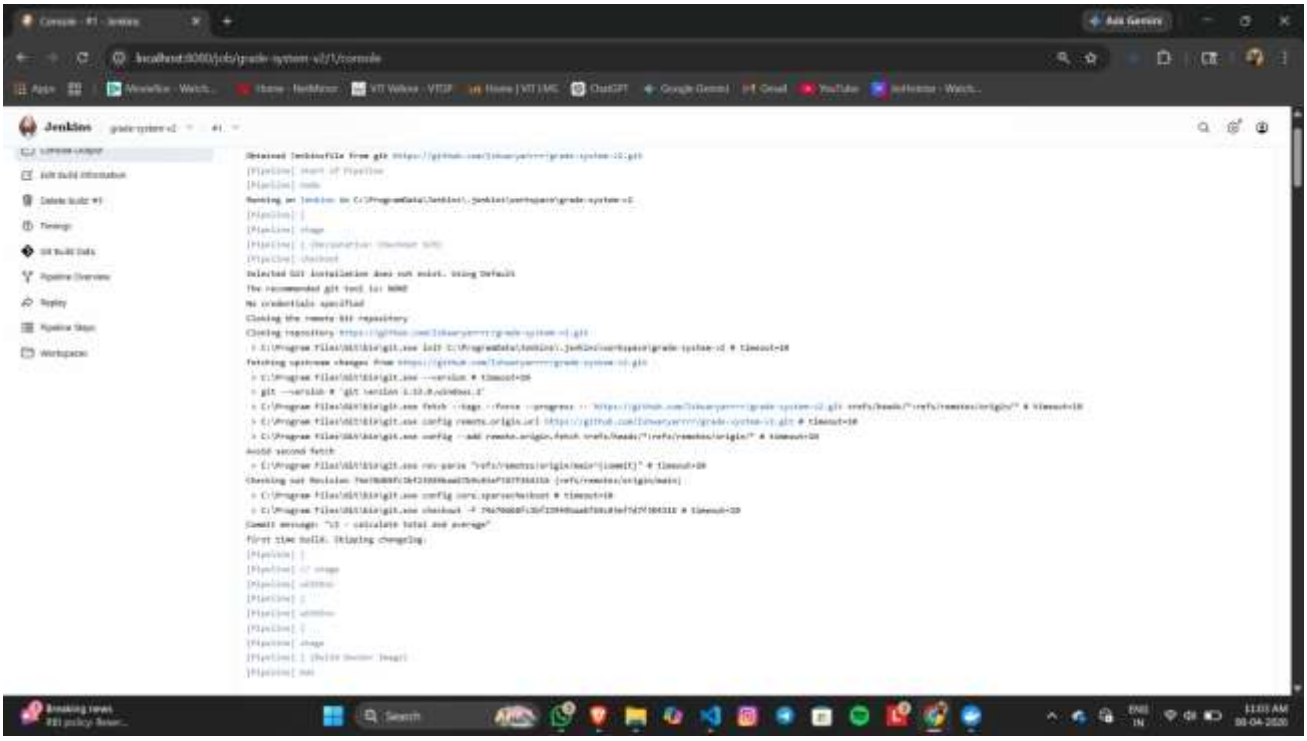
- **Branch Specifier = */main**

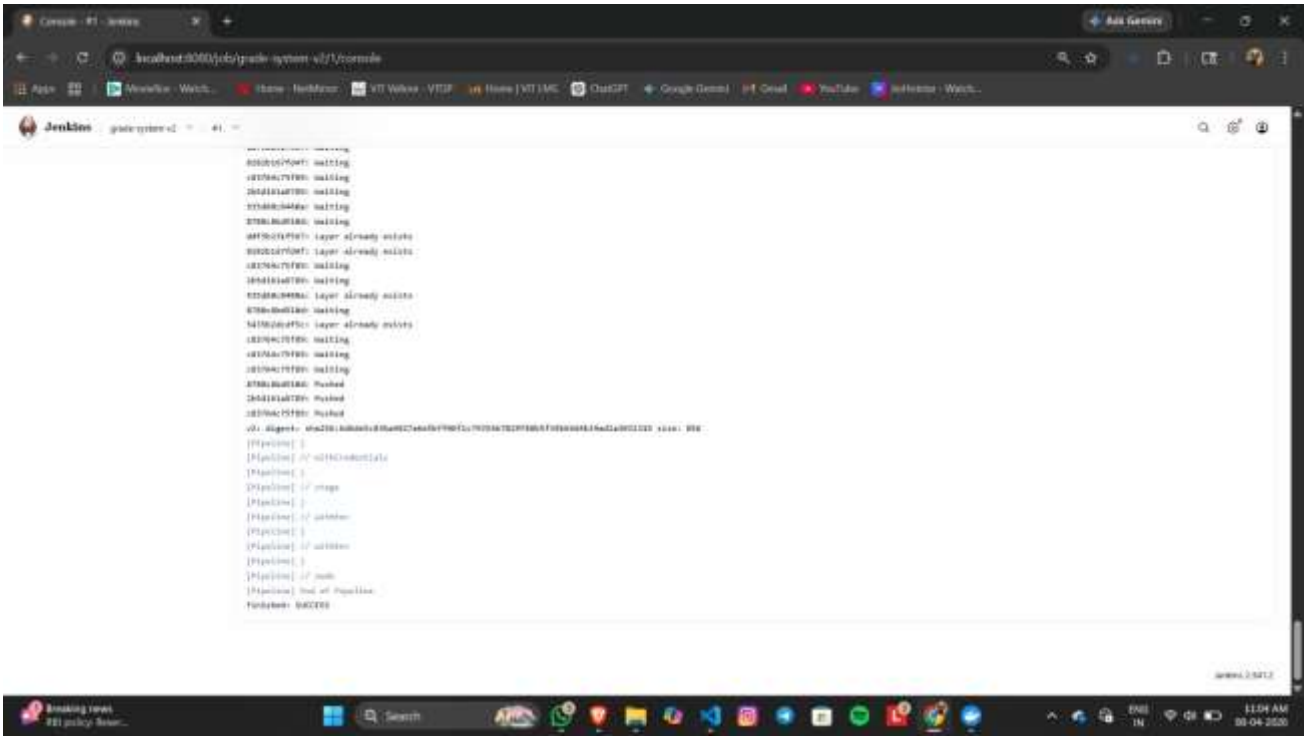


Click: Build Now

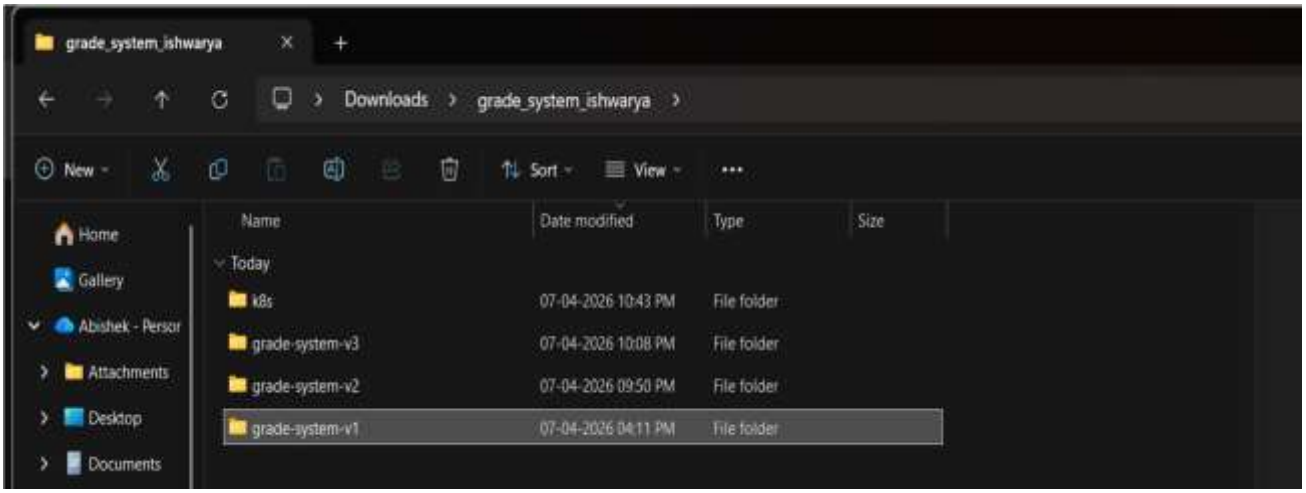


Console Output:

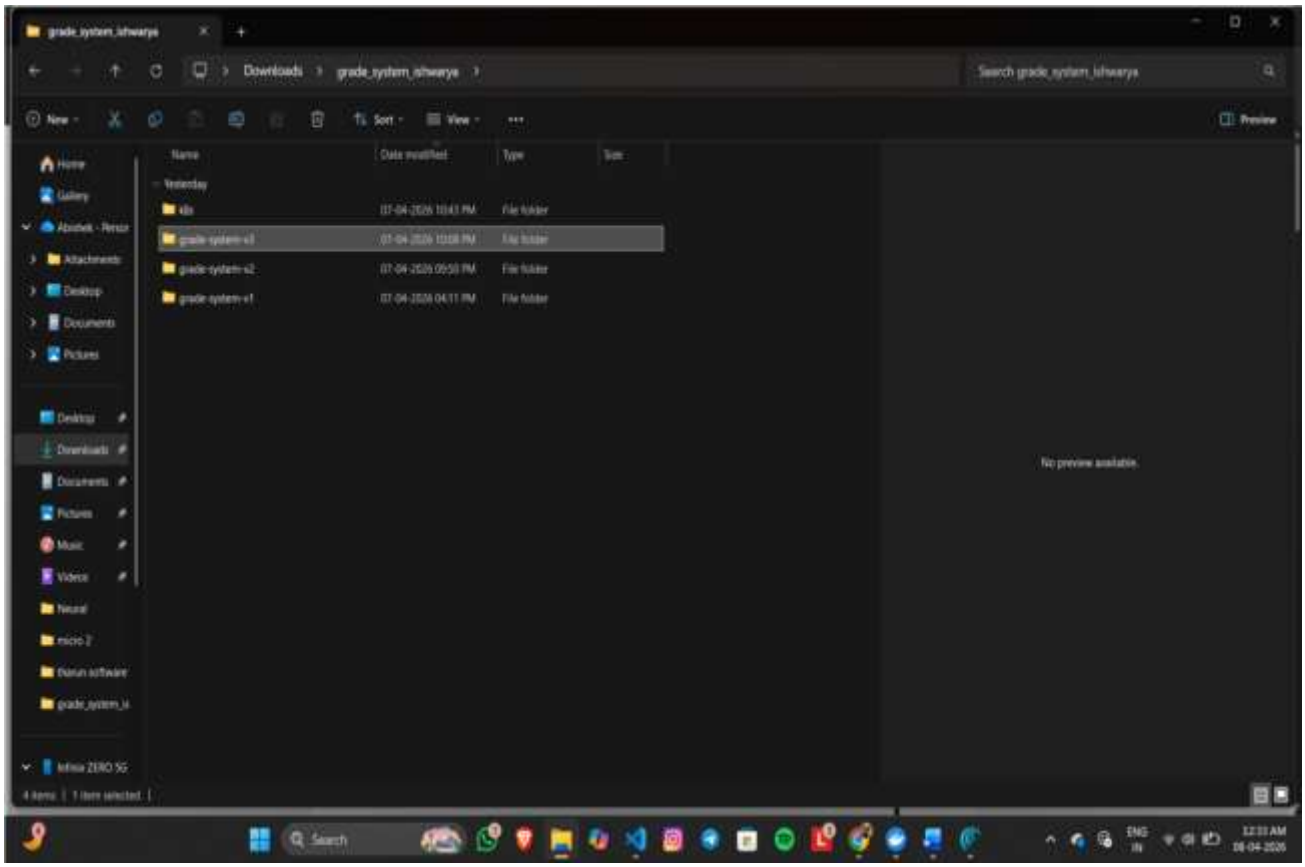




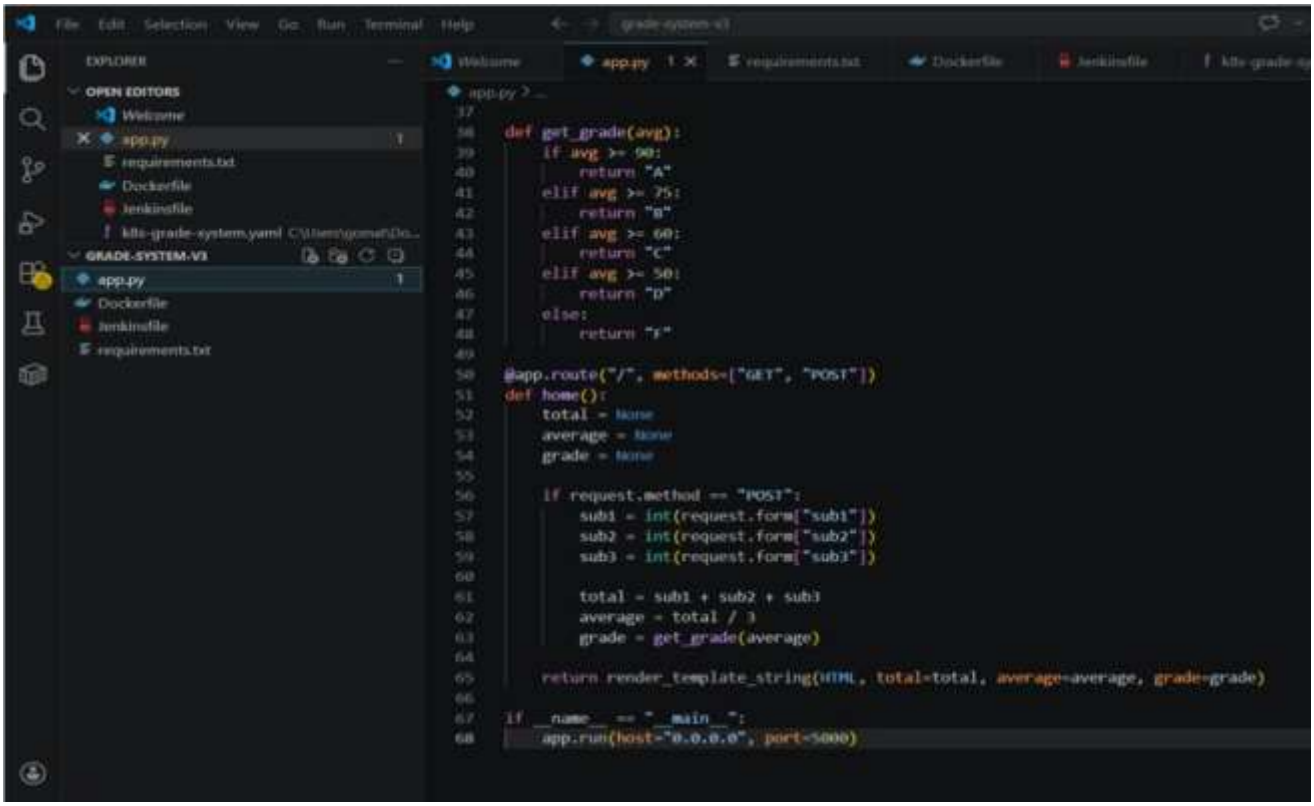
Similarly do it for v3



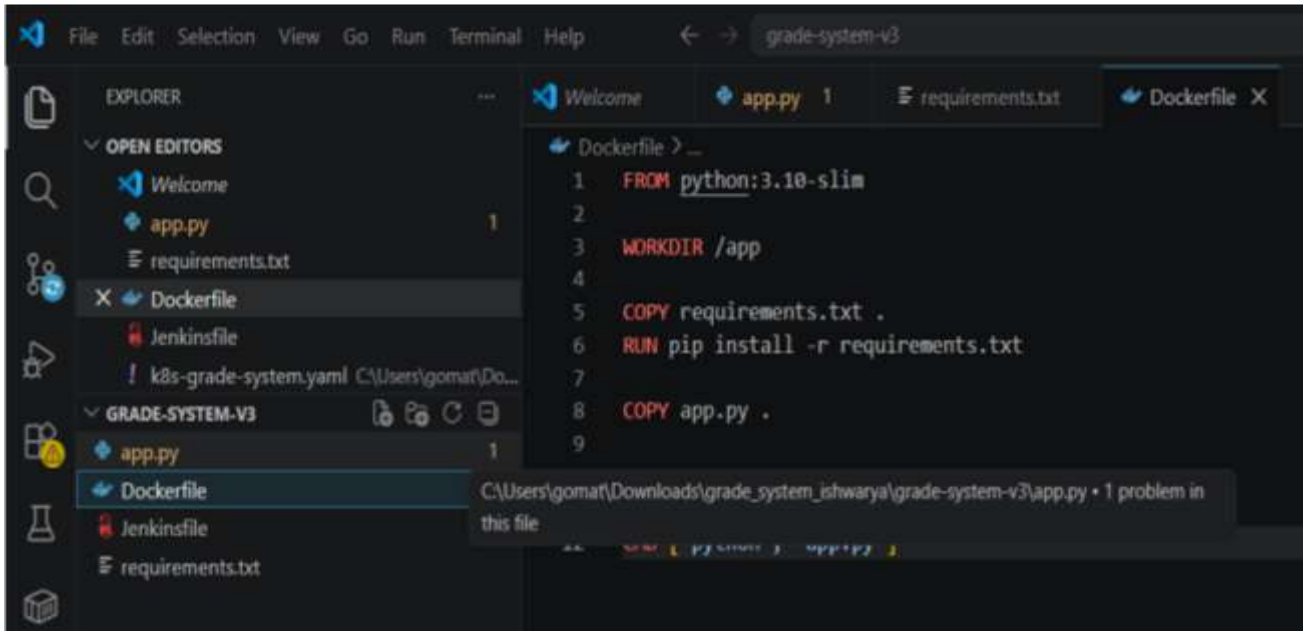
Create a new folder named grade-system-v3 and open it in VS Code



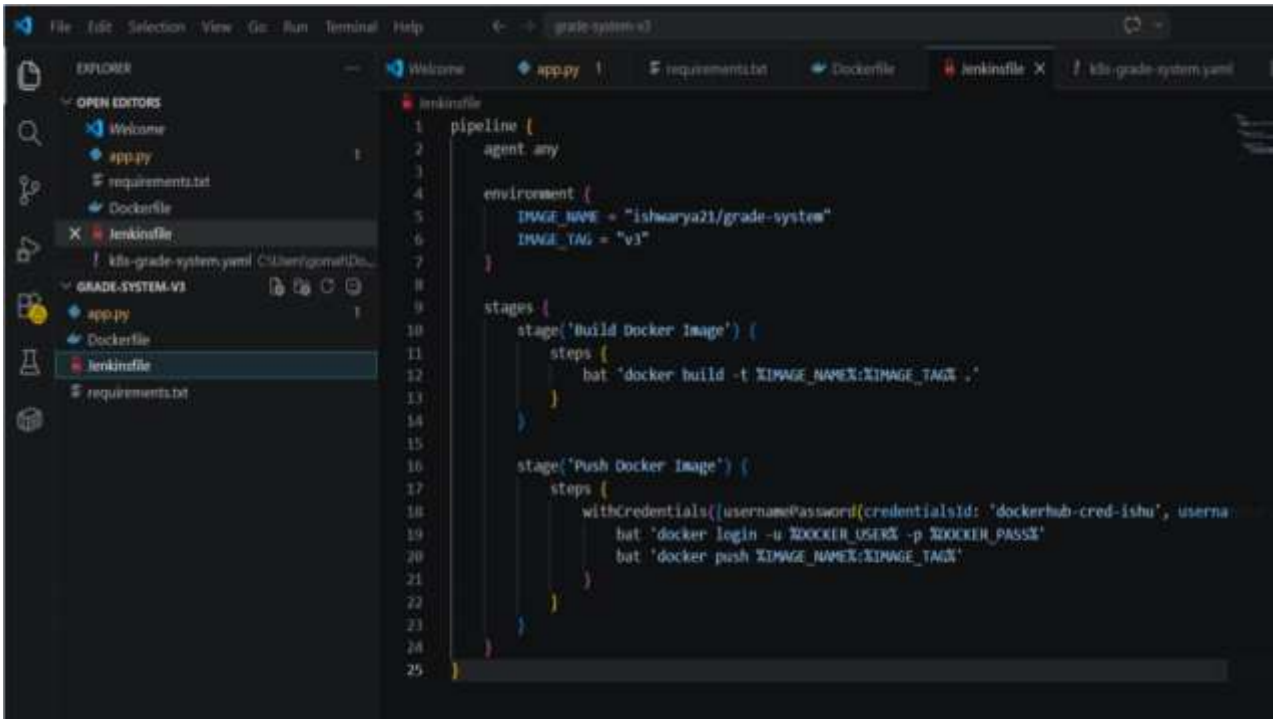
App.py :



Dockerfile:

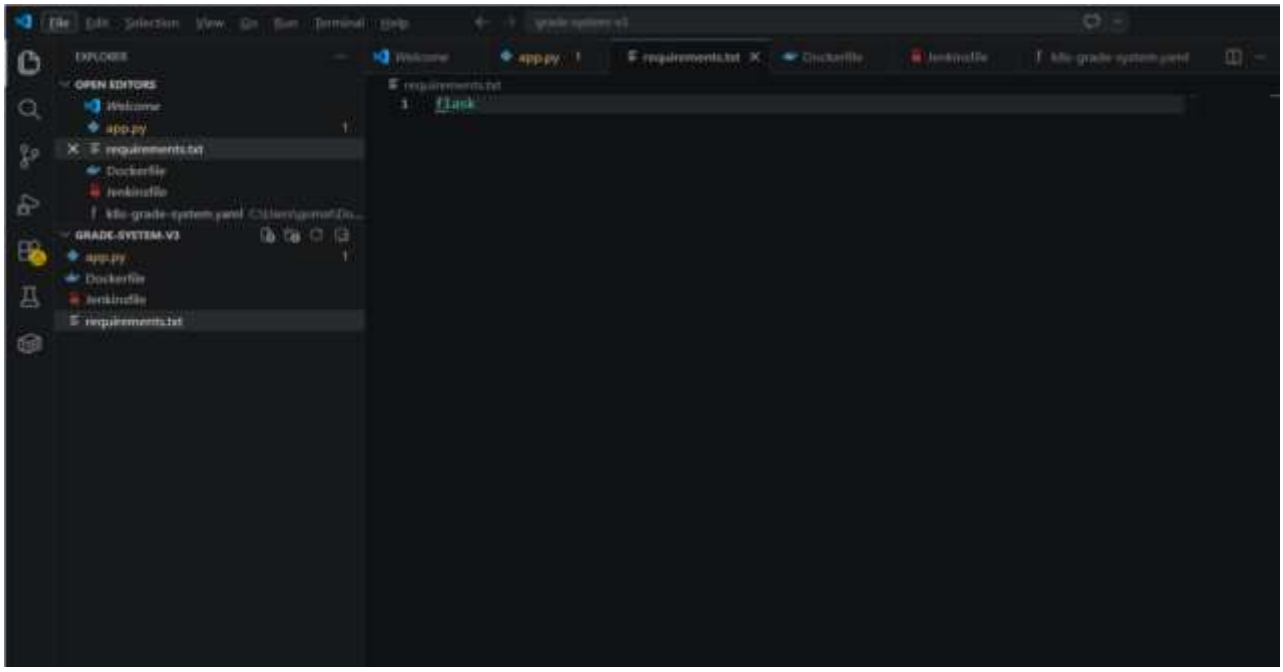


Jenkinsfile:



```
pipeline {
    agent any

    environment {
        IMAGE_NAME = "ishwarya21/grade-system"
        IMAGE_TAG = "v3"
    }

    stages {
        stage('Build Docker Image') {
            steps {
                bat 'docker build -t %IMAGE_NAME%:%IMAGE_TAG% .'
            }
        }

        stage('Push Docker Image') {
            steps {
                withCredentials([usernamePassword(credentialsId: 'dockerhub-cred-ishu', userna
                    bat 'docker login -u %DOCKER_USER% -p %DOCKER_PASS%'
                    bat 'docker push %IMAGE_NAME%:%IMAGE_TAG%'
                }
            }
        }
    }
}
```

Requiement.txt



```
flask
```

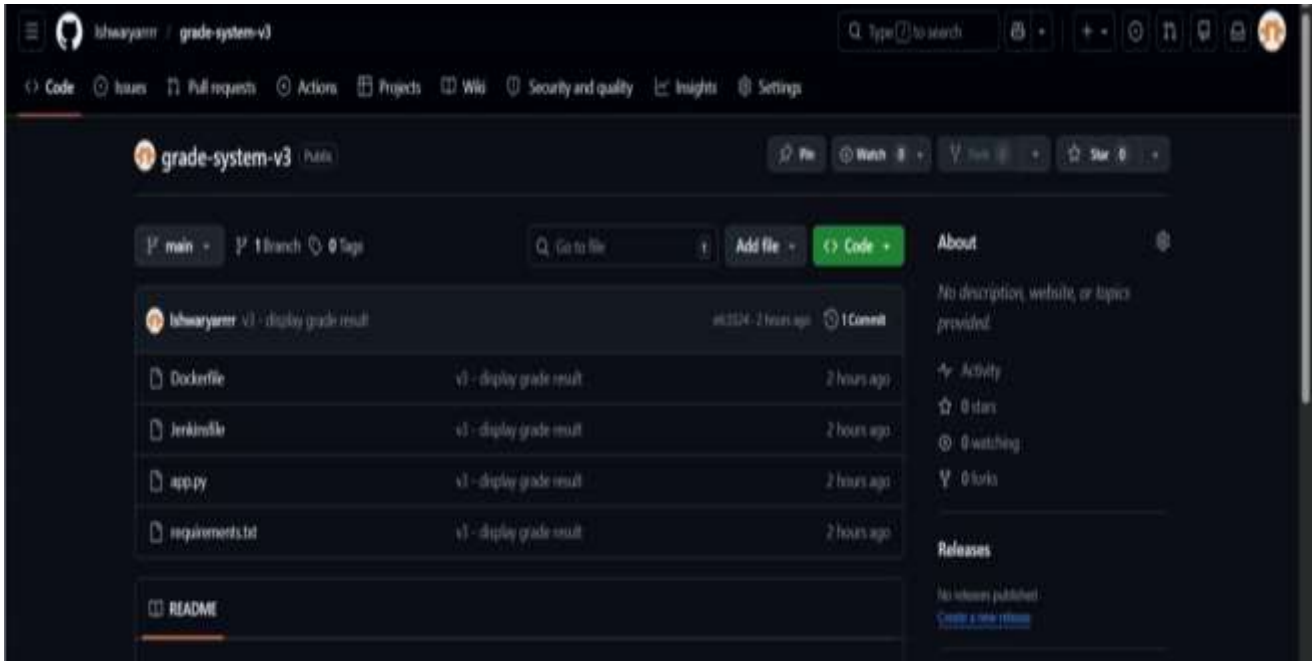
Now push it into github repository:

Create a new repository called grade-system-v3and push it through

git init git add .

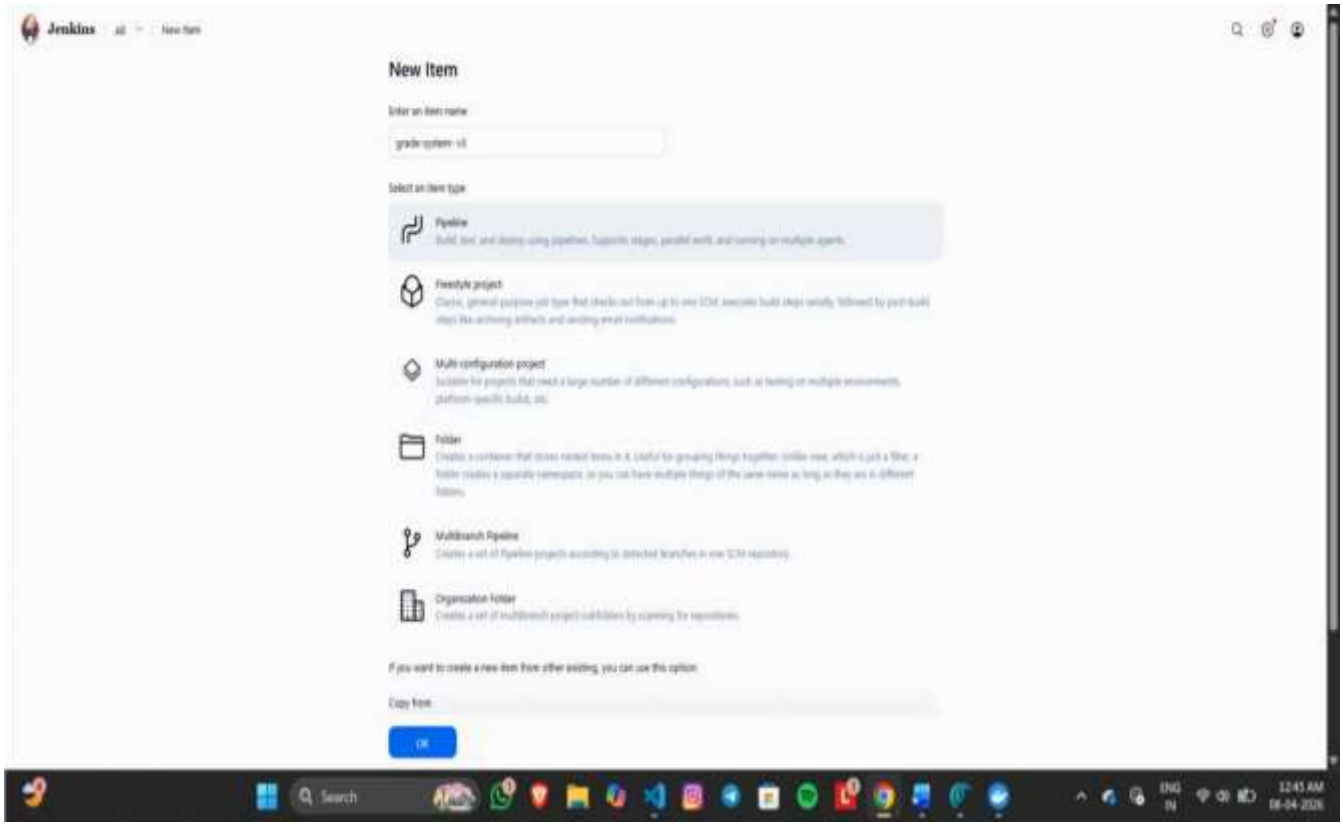
git commit -m "v3 - display grade result" git branch -M main

git remote add origin https://github.com/Ishwaryarrrr/grade-system-v3.git git push -u origin main



Now open Jenkins :

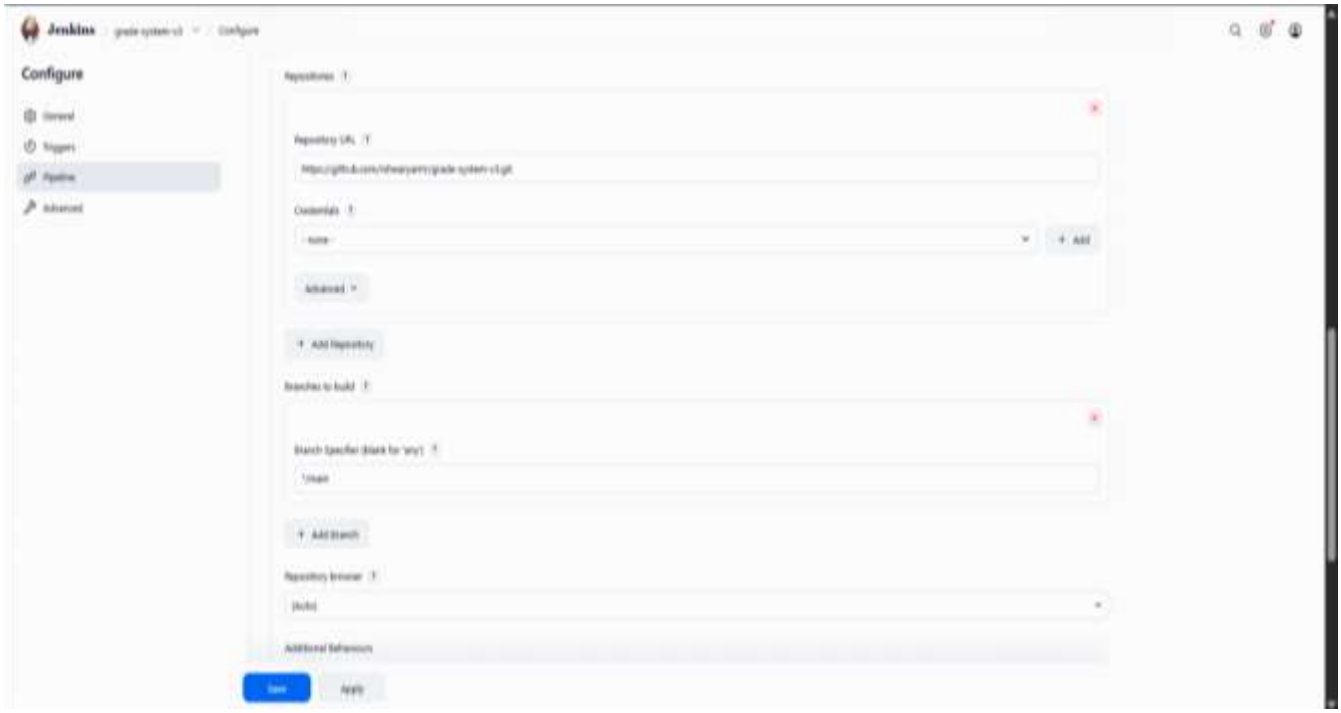
.create new item called garde system v3:



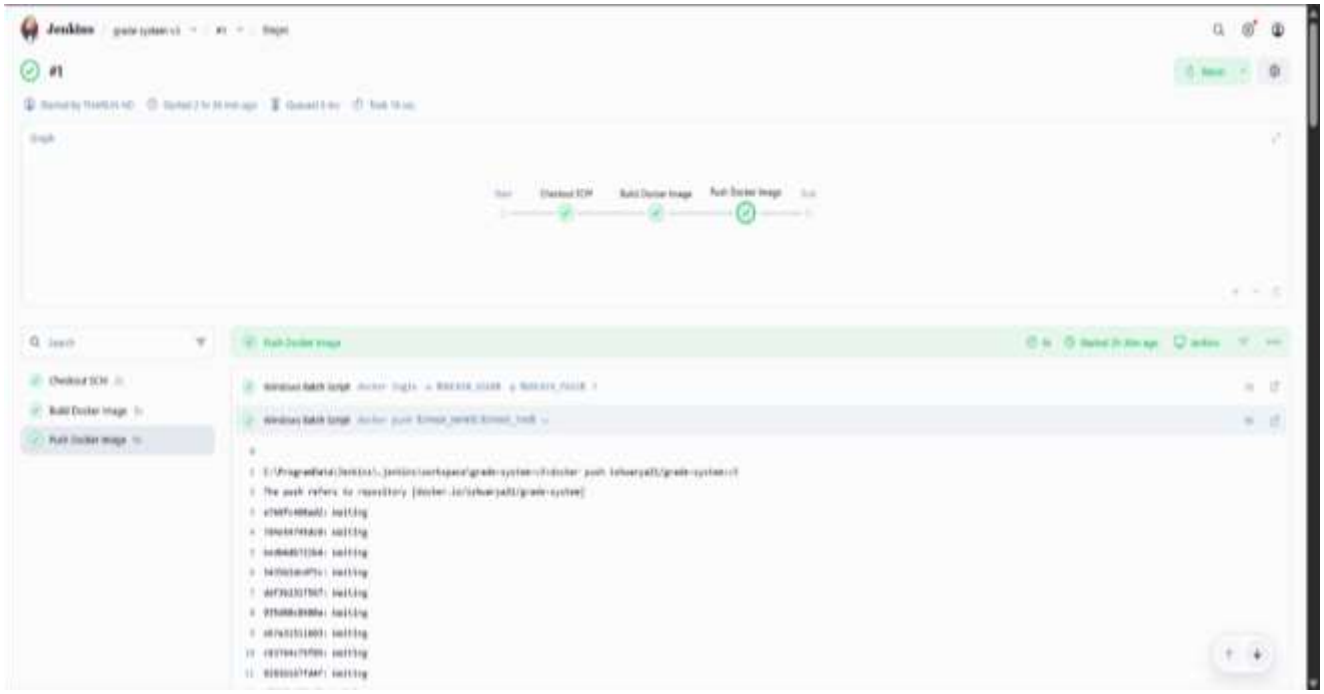
Scroll to Pipeline section.

Set:

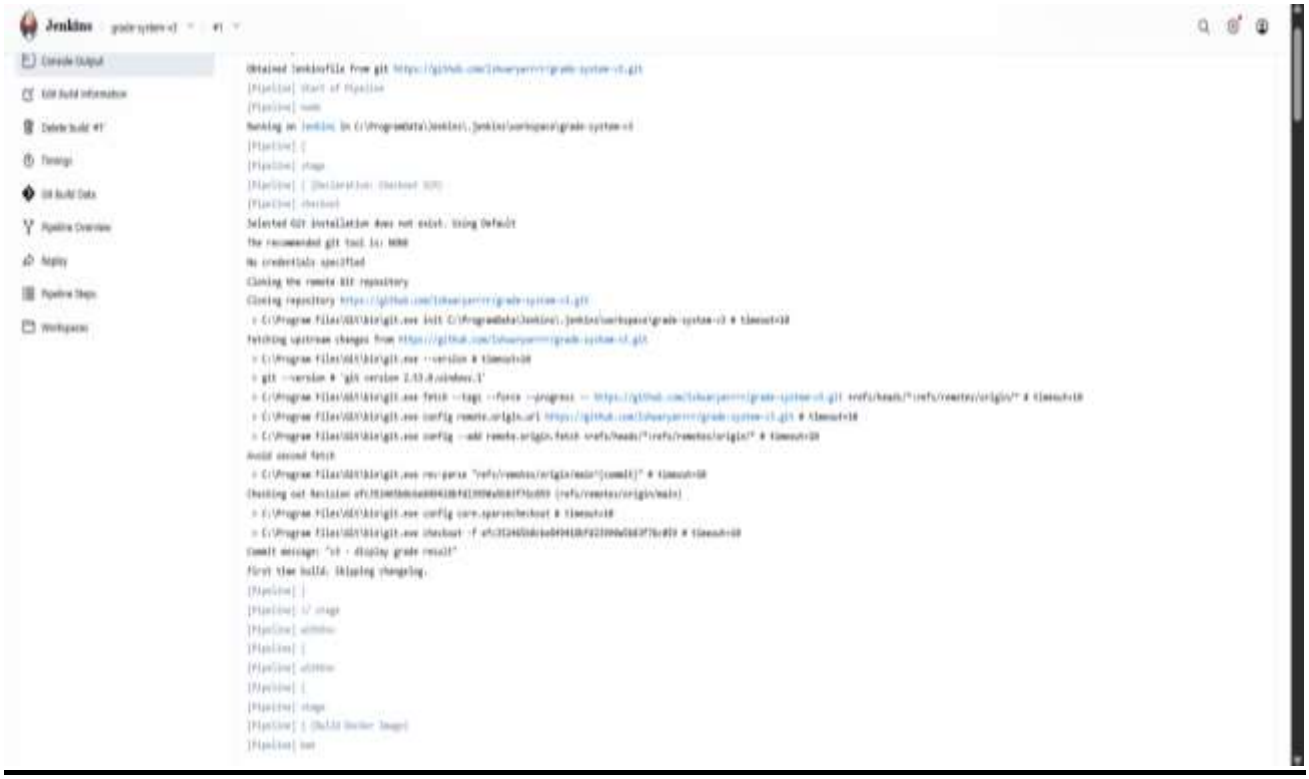
- **SCM =** Git
- **Repository URL =** https://github.com/Ishwaryarrrr/grade-system-v2.git
- **Branch Specifier** = */main



Click: Build Now



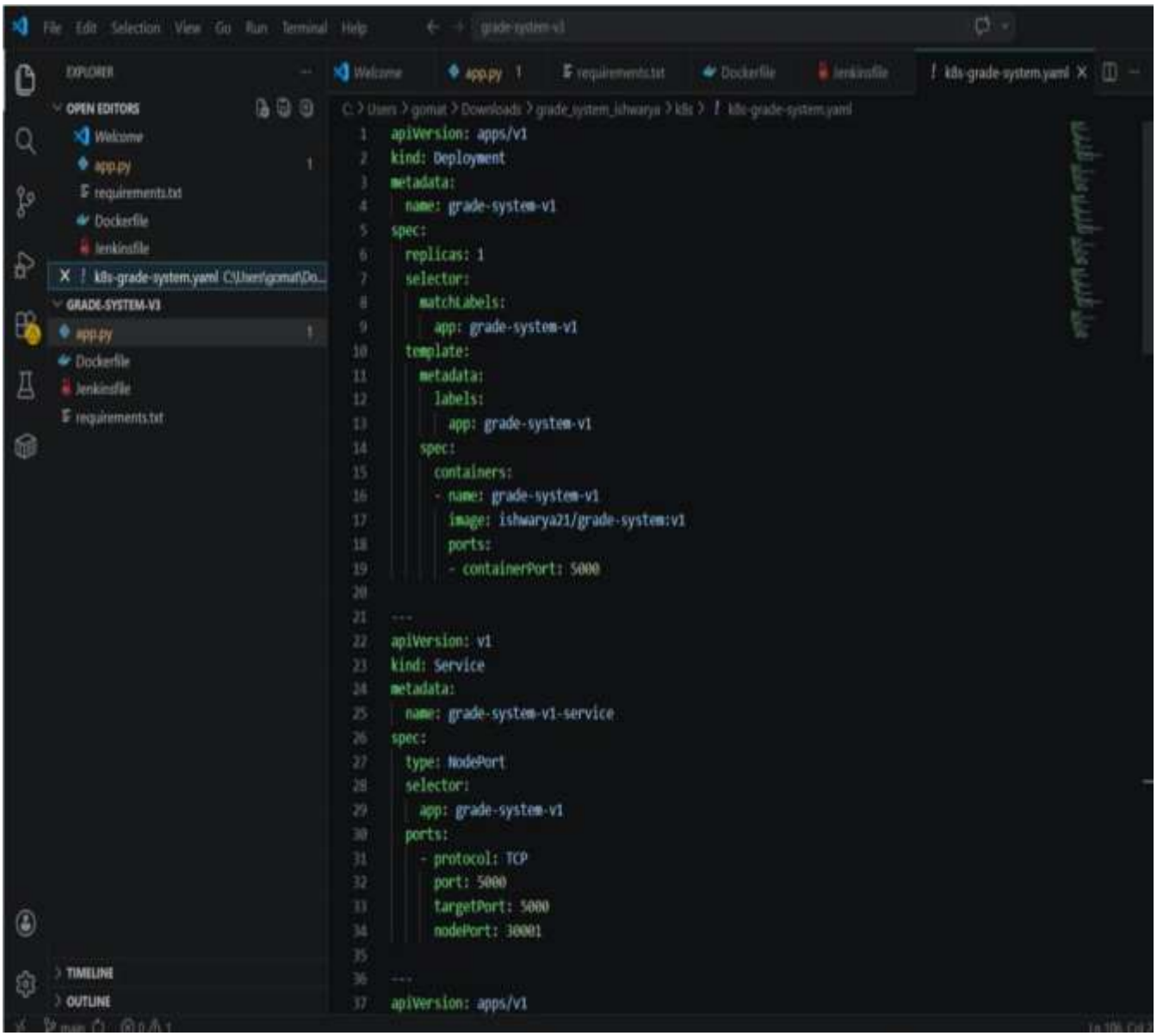
Console Output:



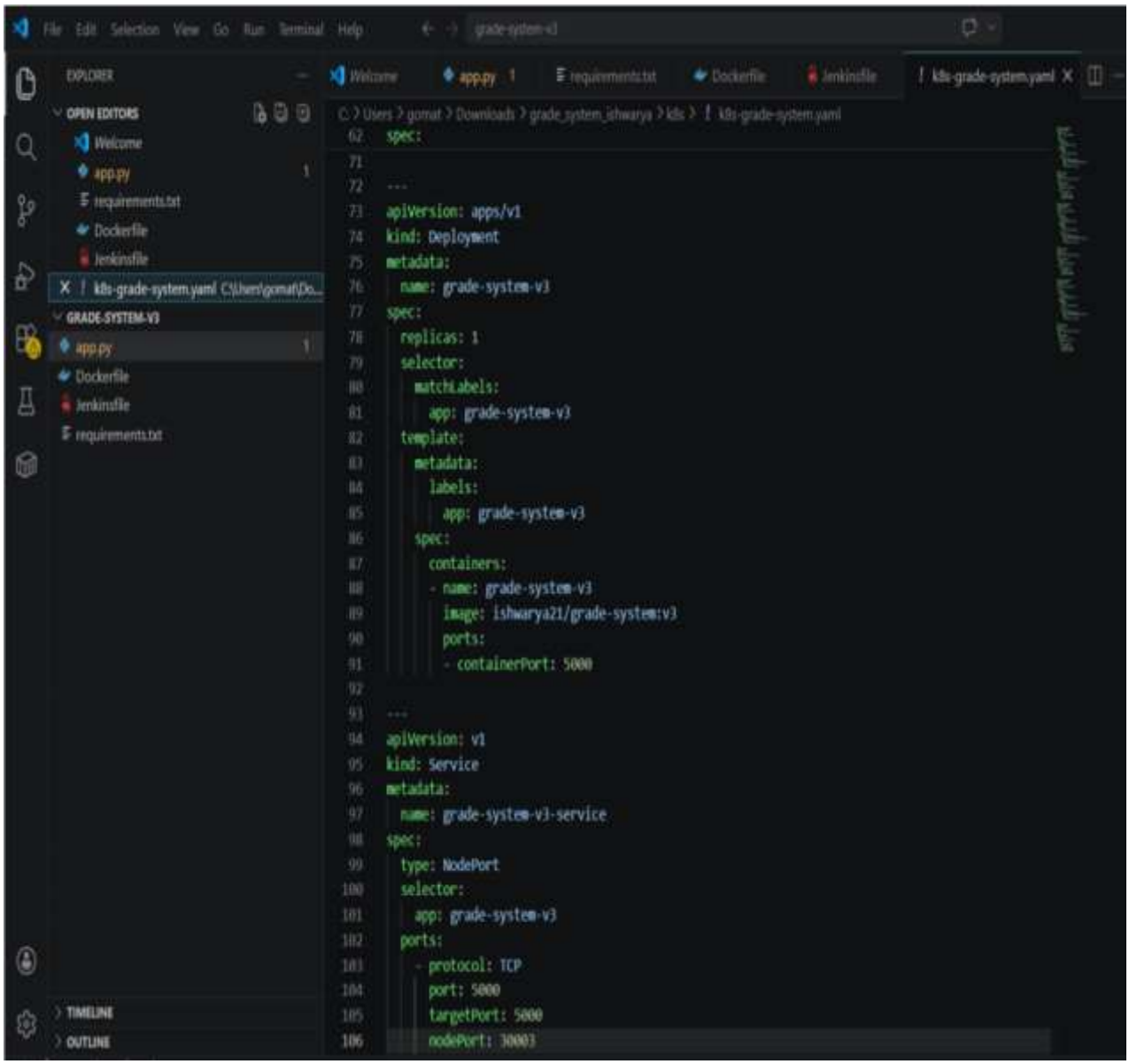
Now we have

- **v1 in GitHub, Jenkins, DockerHub**
- **v2 in GitHub, Jenkins, DockerHub**
- **v3 in GitHub, Jenkins, DockerHub Now the last big part is Kubernetes.**

<u>Create a file named</u>: k8s-grade-system.yaml



```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: grade-system-v1
spec:
  replicas: 1
  selector:
    matchLabels:
      app: grade-system-v1
  template:
    metadata:
      labels:
        app: grade-system-v1
    spec:
      containers:
      - name: grade-system-v1
        image: ishwarya21/grade-system:v1
        ports:
        - containerPort: 5000

---
apiVersion: v1
kind: Service
metadata:
  name: grade-system-v1-service
spec:
  type: NodePort
  selector:
    app: grade-system-v1
  ports:
    - protocol: TCP
      port: 5000
      targetPort: 5000
      nodePort: 30001

---
apiVersion: apps/v1
```

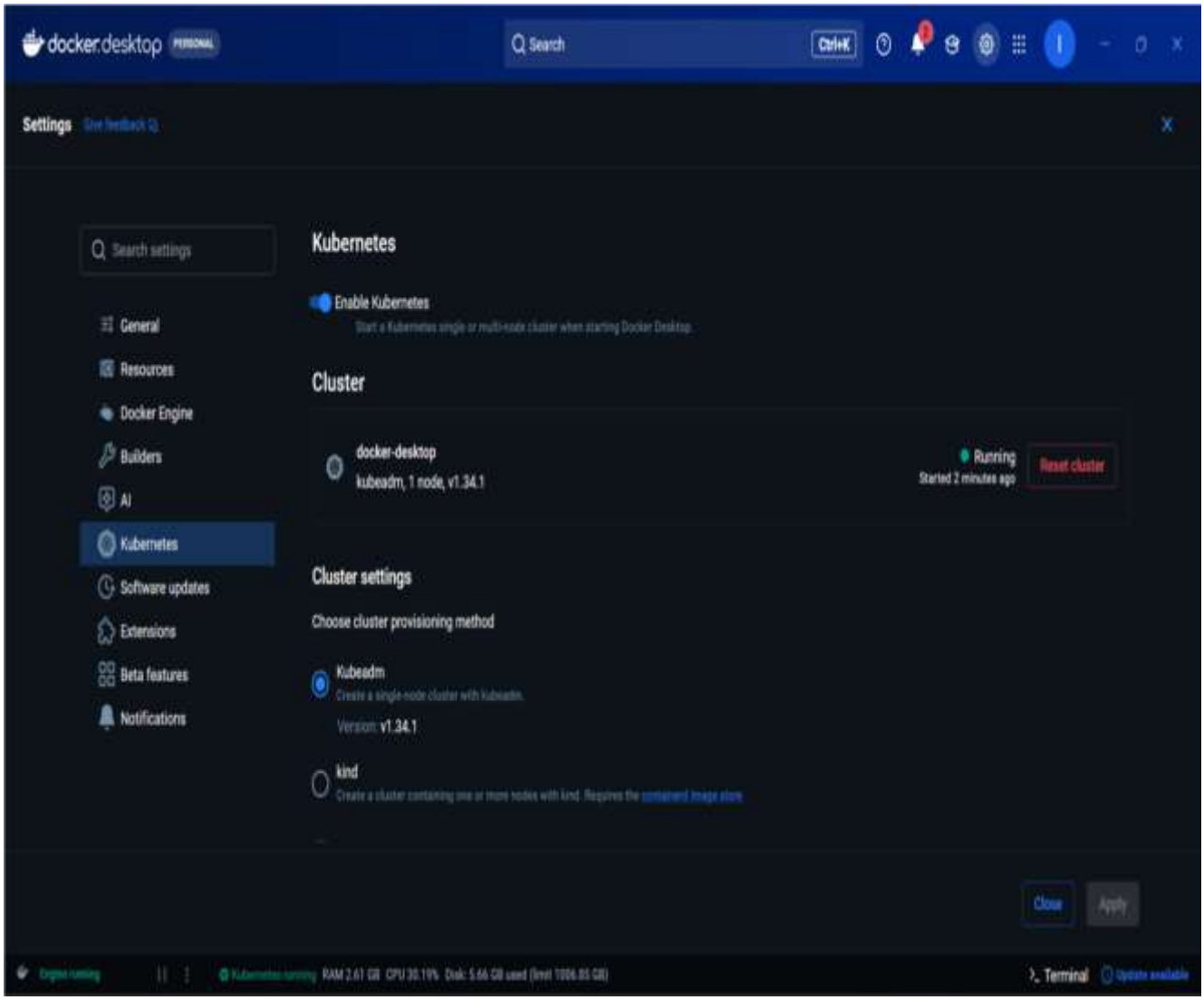


Open Docker Desktop and make sure:

- Docker is running
- Kubernetes is enabled

If Kubernetes is not enabled:

- go to Docker Desktop
- Settings
- Kubernetes
- tick Enable Kubernetes
- Apply and wait until it shows running



Step 1 — Connect to correct cluster

```
C:\Users\gomat>kubectl config use-context docker-desktop
Switched to context "docker-desktop".

C:\Users\gomat>kubectl get nodes
NAME             STATUS   ROLES           AGE    VERSION
docker-desktop   Ready    control-plane   114m   v1.34.1

C:\Users\gomat>
```

Step 2 — Go to your YAML folder

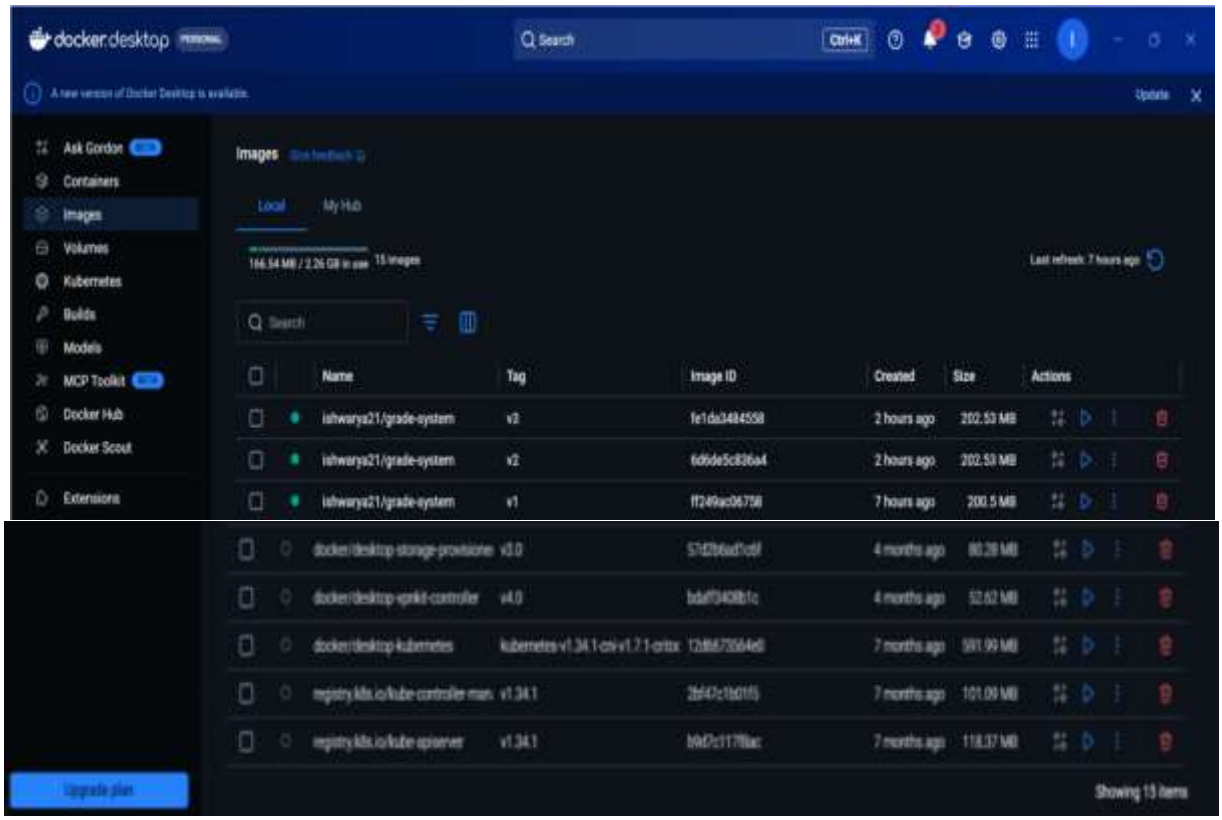
Make sure terminal is inside:C:\Users\gomat\Downloads\grade_system_ishwarya\k8s

Step 3 — Apply again

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\gomat\Downloads\grade_system_ishwarya\k8s>kubectl apply -f k8s-grade-system.yaml
deployment.apps/grade-system-v1 unchanged
service/grade-system-v1-service unchanged
deployment.apps/grade-system-v2 unchanged
service/grade-system-v2-service unchanged
deployment.apps/grade-system-v3 unchanged
service/grade-system-v3-service unchanged

C:\Users\gomat\Downloads\grade_system_ishwarya\k8s>
```

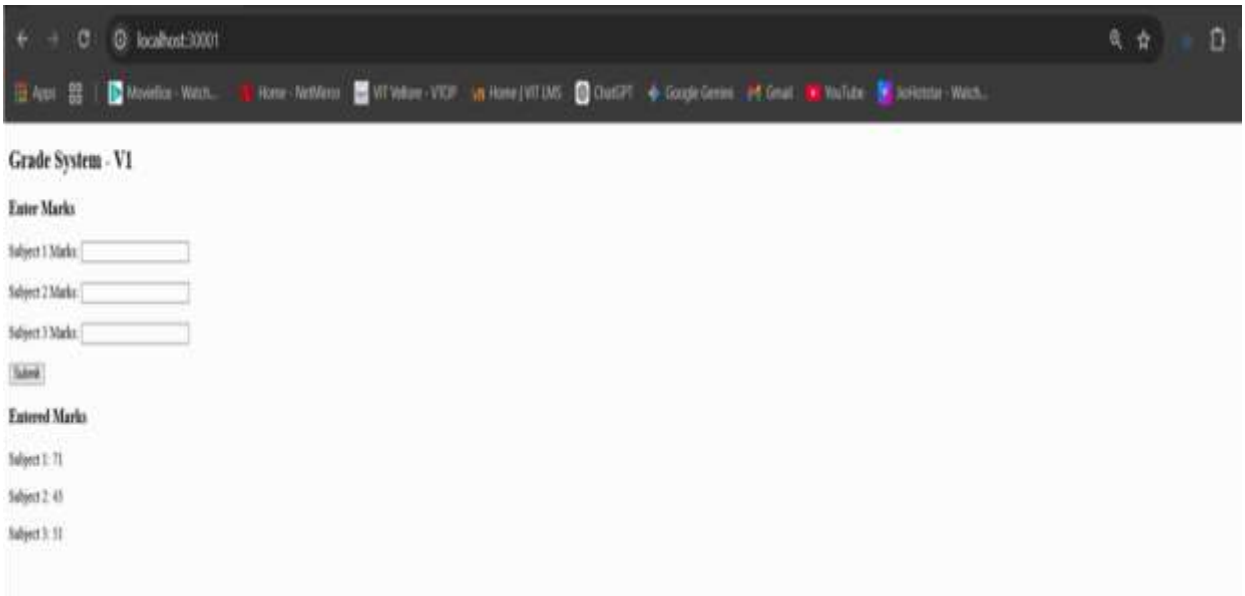
Step 4 — Verify everything

```
C:\Users\gomat\Downloads\grade_system_ishwarya\k8s>kubectl get deployments
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
grade-system-v1   1/1     1            1           101m
grade-system-v2   1/1     1            1           101m
grade-system-v3   1/1     1            1           101m

C:\Users\gomat\Downloads\grade_system_ishwarya\k8s>kubectl get pods
NAME                              READY   STATUS    RESTARTS      AGE
grade-system-v1-5f89868fc8-fkzrb  1/1     Running   1 (62m ago)   101m
grade-system-v2-7c6dddc777-9k9t7  1/1     Running   1 (62m ago)   101m
grade-system-v3-5b6f54474d-lspjq  1/1     Running   1 (62m ago)   101m

C:\Users\gomat\Downloads\grade_system_ishwarya\k8s>kubectl get services
NAME                    TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
grade-system-v1-service NodePort    10.110.208.9     <none>        5000:30001/TCP   101m
grade-system-v2-service NodePort    10.102.47.172    <none>        5000:30002/TCP   101m
grade-system-v3-service NodePort    10.111.192.136   <none>        5000:30003/TCP   101m
kubernetes              ClusterIP   10.96.0.1        <none>        443/TCP          119m
```

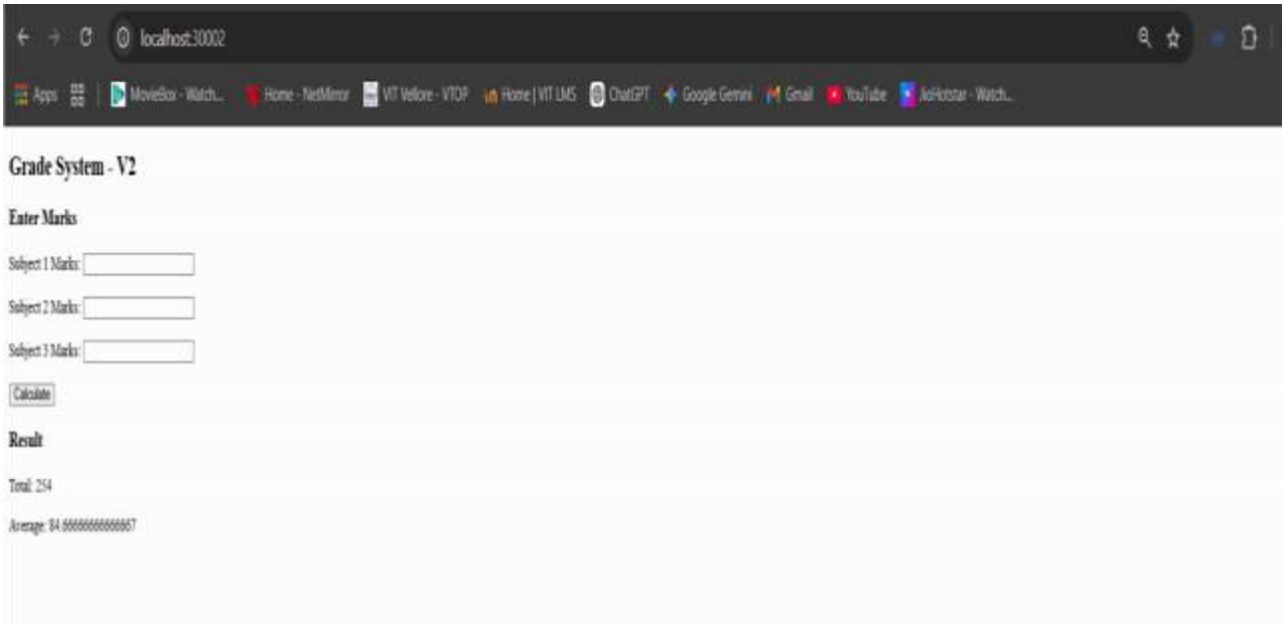
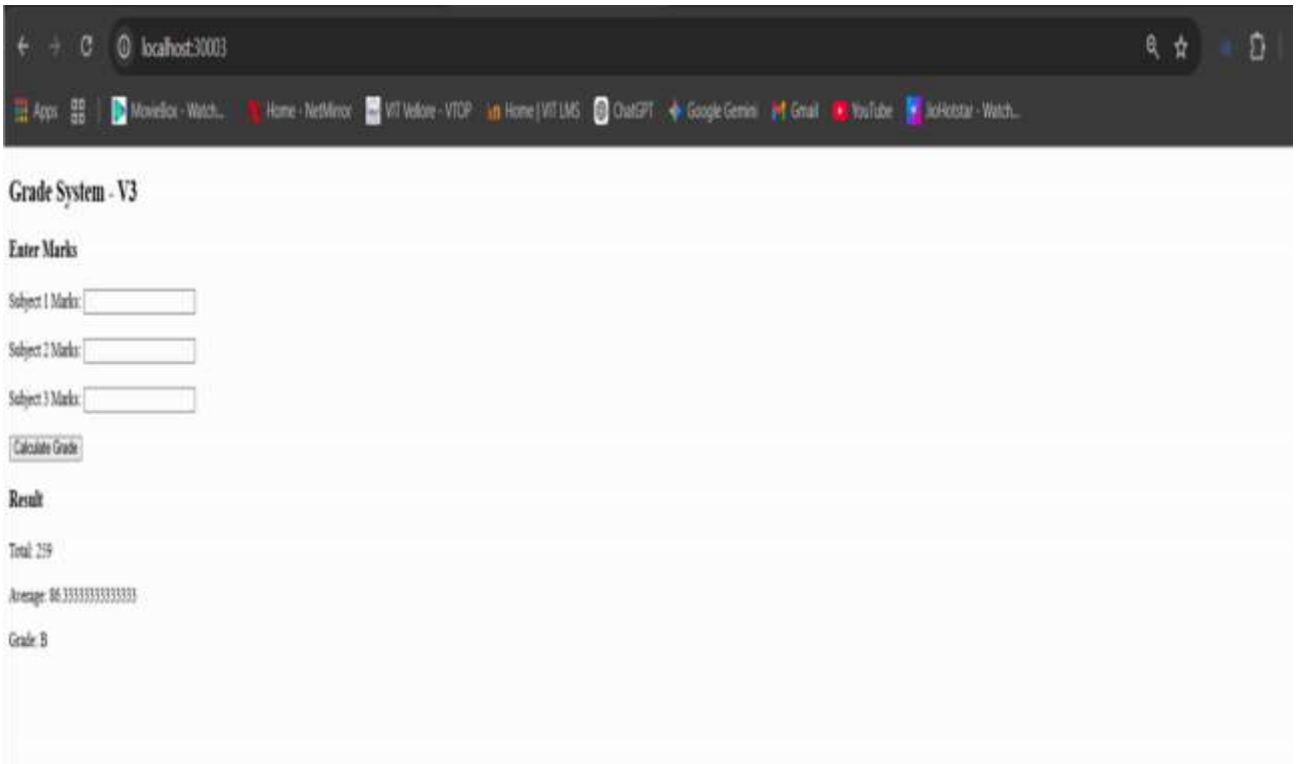


Open browser:

http://localhost:30001



http://localhost:30002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